Padiyar Two-Area Four-Machine System

Power Flow, Small-Signal Stability, and Time-Domain Simulation

13 July 2026

Abstract

This report reconstructs the four-machine, ten-bus, two-area system in Padiyar, *Power System Dynamics: Stability and Control*, second edition, Section 9.6.1. Tables 9.1–9.4 are treated as the primary input source. Each generator uses Padiyar model 1.1, a two-axis transient synchronous-machine model, with either manual excitation or the printed single-time-constant AVR. The AVR case has five states per generator and twenty states in total. The in-house Newton–Raphson power flow, DAE equilibrium, small-signal linearisation and implicit-trapezoidal time-domain simulation are generated from project code. Table 9.5 is used only as a secondary published cross-check; no parameter is fitted to its eigenvalues.

Contents

1 Scope and source discipline

The numerical inputs are transcribed from Padiyar Tables 9.1–9.4 (PDF pages 338–344, printed pages 325–331). The source provides transient-axis parameters X'_d, X'_q, T'_{d0} and T'_{q0}, but no subtransient parameters. Consequently, this report does not invent X'' or T'' data and does not describe the model as sixth order. The two configurations are:

1. **Manual excitation:** four machine states $[\delta, \omega, E'_q, E'_d]$ per generator, 16 states total.

2. **Single-time-constant AVR:** the same machine plus E_{fd}, giving five states per generator and 20 states total.

The cited pages do not provide a PSS design for this case. No PSS gain or time constant is invented; a future PSS comparison requires a separately documented design specification. This case replaces the former use of Kundur Table E12.3 as the report reference.

2 System description

The network contains ten buses and four generators. Generators at buses 1 and 2 form Area 1, while generators at buses 11 and 12 form Area 2. Three parallel circuits connect buses 3 and 13. The total connected active load is 2734 MW on a 100-MVA base. Dynamic loads are represented as constant impedances, as specified by Padiyar.

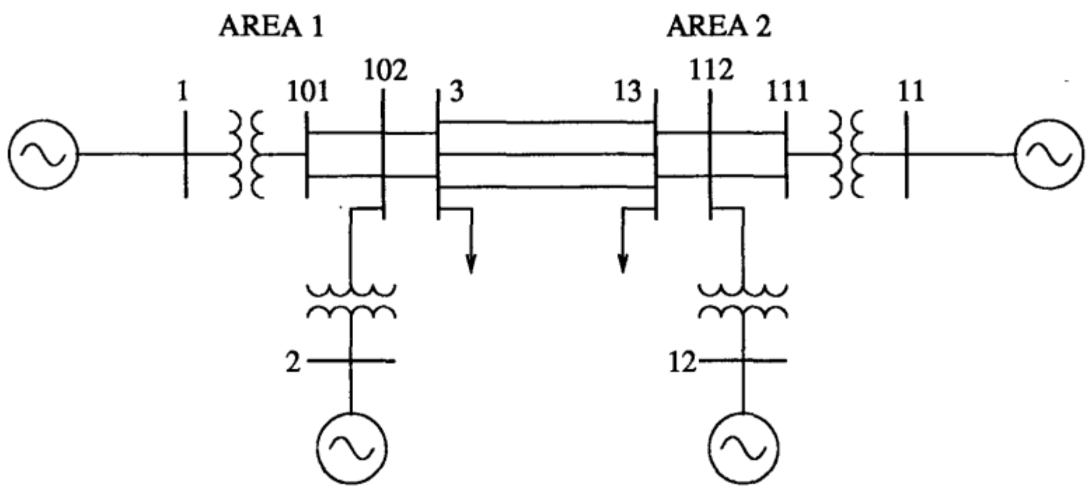

Figure 1: Padiyar four-machine, ten-bus two-area network (source: Padiyar, *Power System Dynamics*, 2nd ed., Sec. 9.6.1). Generators at buses 1 and 2 form Area 1; generators at buses 11 and 12 form Area 2. Three parallel circuits connect buses 3 and 13 across the inter-area tie. Loads are connected at buses 3 and 13.

3 Source data (Padiyar Tables 9.1–9.6)

The following tables are transcribed verbatim from Padiyar, *Power System Dynamics: Stability and Control*, 2nd ed., Section 9.6.1 (printed pages 325–331). They are the *input* to the in-house computation; no value below is a computed result. All quantities are on a 100-MVA base.

Table 1: Transmission line data on 100-MVA base (Padiyar Table 9.1).

From	To	R_s (pu)	X_s (pu)	B (pu)
1	101	0.001	0.012	0.00
2	102	0.001	0.012	0.00
3	13	0.022	0.220	0.33
3	13	0.022	0.220	0.33
3	13	0.022	0.220	0.33
3	102	0.002	0.020	0.03
3	102	0.002	0.020	0.03
11	111	0.001	0.012	0.00
12	112	0.001	0.012	0.00
13	112	0.002	0.020	0.03
13	112	0.002	0.020	0.03
101	102	0.005	0.050	0.075
101	102	0.005	0.050	0.075
111	112	0.005	0.050	0.075
111	112	0.005	0.050	0.075

Table 2: Load-flow data and results — printed operating point (Padiyar Table 9.2).

Bus	$\lvert V\rvert$ (pu)	θ (deg)	P_G (pu)	Q_G (pu)	P_L (pu)	Q_L (pu)	B_{sh} (pu)
1	1.0300	8.2154	7.0	1.3386	0	0	0
2	1.0100	-1.5040	7.0	1.5920	0	0	0
11	1.0300	0.0	7.2172	1.4466	0	0	0
12	1.0100	-10.2051	7.0	1.8083	0	0	0
101	1.0108	3.6615	0	0	0	0	0
102	0.9875	-6.2433	0	0	0	0	0
111	1.0095	-4.6977	0	0	0	0	0
112	0.9850	-14.9443	0	0	0	0	0
3	0.9761	-14.4194	0	0	11.59	2.12	3.0
13	0.9716	-23.2922	0	0	15.75	2.88	4.0

Table 3: Machine data, identical for all four machines unless noted (Padiyar Table 9.3).

Variable	Bus 1	Bus 2	Bus 11	Bus 12
X_l (pu)	0.022	0.022	0.022	0.022
R_a (pu)	0.00028	0.00028	0.00028	0.00028
X_d (pu)	0.2	0.2	0.2	0.2
X'_d (pu)	0.033	0.033	0.033	0.033
T'_{d0} (s)	8.0	8.0	8.0	8.0
X_q (pu)	0.19	0.19	0.19	0.19
X'_q (pu)	0.061	0.061	0.061	0.061
T'_{q0} (s)	0.4	0.4	0.4	0.4
H (s)	54.0	54.0	63.0	63.0
D (pu)	0	0	0	0

Table 4: Excitation system data, single-time-constant AVR (Padiyar Table 9.4).

Variable	Bus 1	Bus 2	Bus 11	Bus 12
K_A (pu)	200	200	200	200
T_A (s)	0.02	0.02	0.02	0.02

Table 5: Participation factors of the swing modes (Padiyar Table 9.6). ($\omega_1 = 7.29$, $\omega_2 = 6.69$, $\omega_3 = 4.44$ rad/s).

Variable	Mode #1	Mode #2	Mode #3
ΔS_{m1} (bus 1)	$-0.18 \pm j0.14$	$0.004 \pm j0.001$	$-0.06 \pm j0.15$
ΔS_{m2} (bus 2)	$-0.27 \pm j0.14$	$0.00 \pm j0.00$	$-0.062 \pm j0.07$
ΔS_{m3} (bus 11)	$-0.00 \pm j0.00$	$0.24 \pm j0.03$	$-0.068 \pm j0.13$
ΔS_{m4} (bus 12)	$-0.04 \pm j0.00$	$0.30 \pm j0.08$	$-0.061 \pm j0.07$

The inter-area mode (mode 3, $\omega \approx 4.44$ rad/s) has comparable participation from all four machines, confirming its inter-area character. Modes 1 and 2 are local to Area 1 and Area 2 respectively.

4 Power-flow analysis

The project-owned Newton–Raphson solver uses bus 11 as the reference bus, matching the zero angle printed in Table 9.2. Buses 1, 2 and 12 are PV buses. Line susceptance B from Table 9.1 is represented as $B/2$ at each end of the standard π model. This convention reproduces the printed operating point.

4.1 Newton–Raphson formulation

The solver iterates on the polar voltage state $x = [\theta_2, \ldots, \theta_N, |V|_{PQ_1}, \ldots]^T$ (slack angle fixed, PV magnitudes fixed). Bus power injections are computed from the admittance matrix $Y = G + jB$:

$$P_i(x) = |V_i| \sum_{j=1}^{N} |V_j| \Big(G_{ij} \cos\theta_{ij} + B_{ij} \sin\theta_{ij} \Big), \tag{1}$$

$$Q_i(x) = |V_i| \sum_{j=1}^{N} |V_j| \Big(G_{ij} \sin\theta_{ij} - B_{ij} \cos\theta_{ij} \Big), \tag{2}$$

where $\theta_{ij} = \theta_i - \theta_j$. The power mismatch is $\Delta P_i = P_i^{\text{spec}} - P_i(x)$, $\Delta Q_i = Q_i^{\text{spec}} - Q_i(x)$. At each iteration the Newton step solves

$$J(x_k)\,\Delta x = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix}, \qquad x_{k+1} = x_k + \Delta x, \tag{3}$$

where J is the standard 4-block Jacobian with sub-matrices $H = \partial P/\partial\theta$, $N = \partial P/\partial|V|$, $M = \partial Q/\partial\theta$, $L = \partial Q/\partial|V|$. Iteration terminates when $\max|\Delta P, \Delta Q| < 10^{-10}$. No external nonlinear solver is used; the Jacobian is built analytically from Y in project code.

Table 6: Power-flow convergence and Table 9.2 comparison.

Quantity	Value
Converged	Yes
Iterations	3
Maximum $\|\Delta V\|$ vs Table 9.2	4.927e-05 pu
Maximum $\|\Delta\theta\|$ vs Table 9.2	4.983e-05 deg
Total generation	2821.72 MW
Total load	2734.00 MW
Total loss	87.72 MW

Table 7: Power-flow comparison: Padiyar Table 9.2 versus the in-house solution. Absolute differences are reported because an angle percentage error is undefined at the zero-angle reference bus.

Bus	V_{book} (pu)	V_{ours} (pu)	$\|\Delta V\|$ (pu)	θ_{book} (deg)	θ_{ours} (deg)	$\|\Delta\theta\|$ (deg)
1	1.0300	1.0300	0.00e+00	8.2154	8.2154	4.98e-05
2	1.0100	1.0100	0.00e+00	-1.5040	-1.5040	1.13e-05
11	1.0300	1.0300	0.00e+00	0.0000	0.0000	0.00e+00
12	1.0100	1.0100	0.00e+00	-10.2051	-10.2051	6.79e-06
101	1.0108	1.0108	5.72e-07	3.6615	3.6615	1.83e-05
102	0.9875	0.9875	3.06e-05	-6.2433	-6.2433	2.33e-05
111	1.0095	1.0095	3.02e-05	-4.6977	-4.6977	3.44e-05
112	0.9850	0.9850	4.83e-05	-14.9443	-14.9443	4.15e-05
3	0.9761	0.9761	1.43e-05	-14.4194	-14.4194	3.67e-05
13	0.9716	0.9716	4.93e-05	-23.2922	-23.2922	1.74e-05

Table 8: Detailed in-house power-flow results (mirrors the launcher output). All powers are in pu on the 100-MVA base.

Bus	Type	$\lvert V\rvert$ (pu)	θ (deg)	P_G (pu)	Q_G (pu)	P_L (pu)	Q_L (pu)	P_{net} (pu)	Q_{net} (pu)
1	PV	1.0300	8.2154	7.0000	1.3386	0.0000	0.0000	7.0000	1.3386
2	PV	1.0100	-1.5040	7.0000	1.5920	0.0000	0.0000	7.0000	1.5920
11	REF	1.0300	0.0000	7.2172	1.4466	0.0000	0.0000	7.2172	1.4466
12	PV	1.0100	-10.2051	7.0000	1.8083	0.0000	0.0000	7.0000	1.8083
101	PQ	1.0108	3.6615	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
102	PQ	0.9875	-6.2433	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
111	PQ	1.0095	-4.6977	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
112	PQ	0.9850	-14.9443	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	PQ	0.9761	-14.4194	0.0000	0.0000	11.5900	2.1200	-11.5900	-2.1200
13	PQ	0.9716	-23.2922	0.0000	0.0000	15.7500	2.8800	-15.7500	-2.8800

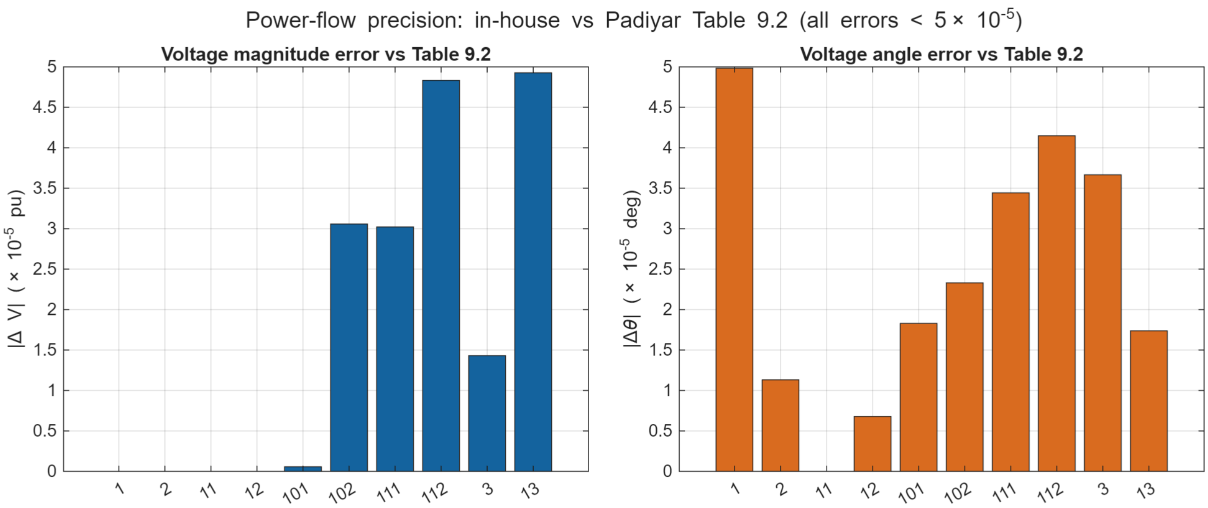

Figure 2: Power-flow precision: in-house values versus Padiyar Table 9.2. Top row—voltage-magnitude and angle scatter with the 1:1 diagonal (points on the line indicate exact agreement). Bottom row—per-bus absolute errors $|\Delta V|$ and $|\Delta\theta|$, all below 5×10^{-5}.

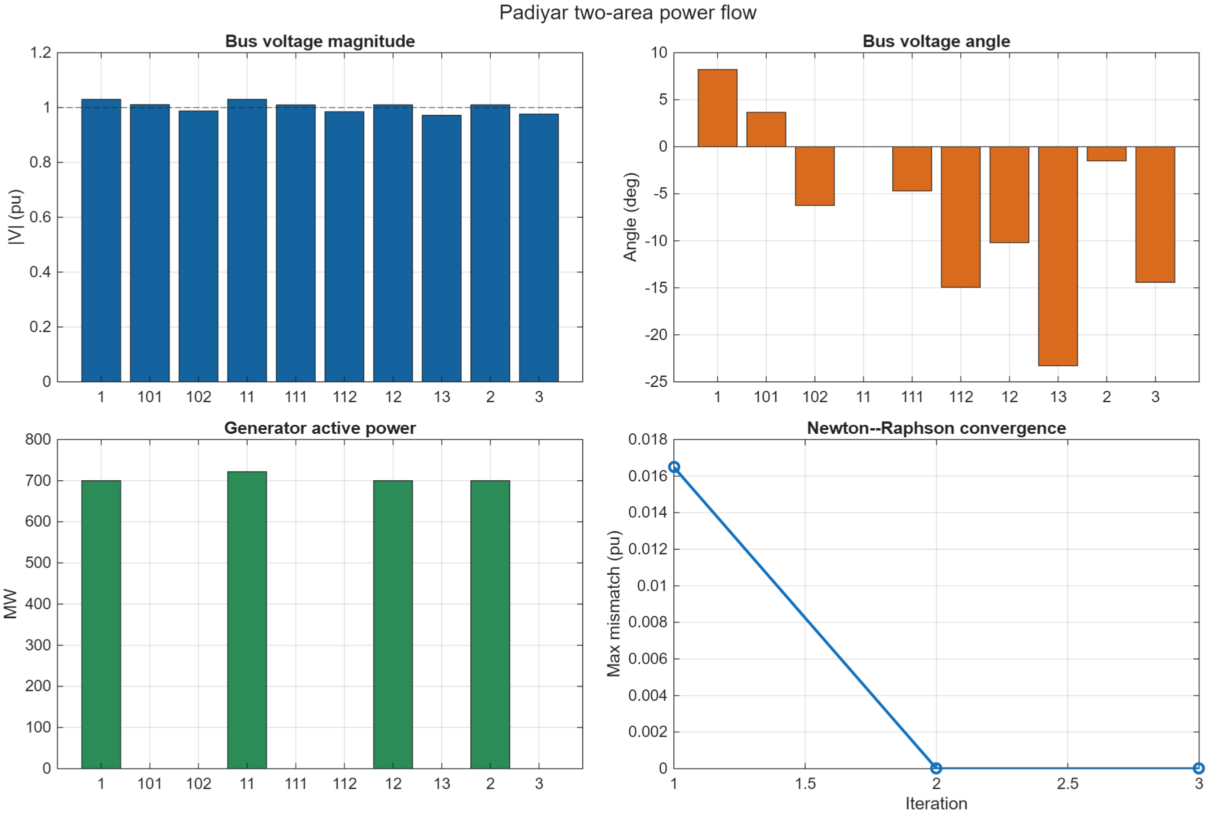

Figure 3: In-house power-flow result and Newton–Raphson convergence.

5 Model 1.1 with AVR

For generator i, the state order and its corresponding derivative order are

$$x_i = \left[\delta_i,\ \omega_i,\ E'_{qi},\ E'_{di},\ E_{fdi}\right]^T, \qquad \dot{x}_i = \left[\dot{\delta}_i,\ \dot{\omega}_i,\ \dot{E}'_{qi},\ \dot{E}'_{di},\ \dot{E}_{fdi}\right]^T = f_i(x_i, I_{di}, I_{qi}, |V_i|, u_i), \tag{4}$$

where the overdot denotes the time derivative. The complete input-driven DAE is

$$\dot{x} = f(x, y, u), \qquad 0 = g(x, y, u), \tag{5}$$

where

$$\begin{aligned} y &= \left[\Re(V)^T \quad \Im(V)^T\right]^T, \qquad u = \left[P_m^T \quad u_{exc}^T \quad \rho\right]^T, \\ u_{exc} &= \begin{cases} V_{ref}, & \text{AVR}, \\ E_{fd0}, & \text{manual excitation}. \end{cases} \end{aligned} \tag{6}$$

Thus x contains differential machine/AVR states, y contains algebraic bus voltages, and u contains prescribed inputs: mechanical power P_m, excitation command u_{exc}, and topology selector ρ. Equilibrium and SSSA hold $u = u_0$ fixed; TS changes only $\rho(t)$ at the declared event times. Generator current is positive from the machine into the network. The dynamic equations implemented in the shared DAE are

$$\dot{\delta}_i = \omega_B(\omega_i - 1), \tag{7}$$
$$2H_i\dot{\omega}_i = P_{mi} - P_{ei} - D_i(\omega_i - 1), \tag{8}$$
$$T'_{d0i}\dot{E}'_{qi} = E_{fdi} - E'_{qi} - (X_{di} - X'_{di})I_{di}, \tag{9}$$
$$T'_{q0i}\dot{E}'_{di} = -E'_{di} + (X_{qi} - X'_{qi})I_{qi}, \tag{10}$$
$$T_{Ai}\dot{E}_{fdi} = K_{Ai}(V_{ref,i} - |V_i|) - E_{fdi}. \tag{11}$$

The transient-saliency stator interface is

$$V_{di} = E'_{di} - R_aI_{di} + X'_qI_{qi}, \tag{12}$$
$$V_{qi} = E'_{qi} - R_aI_{qi} - X'_dI_{di}. \tag{13}$$

Air-gap electrical power is calculated directly as

$$P_{ei} = V_{di}I_{di} + V_{qi}I_{qi} + R_a(I_{di}^2 + I_{qi}^2). \tag{14}$$

The same nonlinear functions are used for equilibrium, SSSA and TS.

5.1 AVR versus manual excitation

Two excitation configurations are studied from the same DAE:

- **With AVR (5 states/machine, 20 total).** The field voltage E_{fd} is a dynamic state governed by the single-time-constant AVR of Table 9.4 ($K_A = 200$, $T_A = 0.02$ s):

  $$T_A\dot{E}_{fd} = K_A(V_{ref} - |V|) - E_{fd}. \tag{15}$$

  At equilibrium $\dot{E}_{fd} = 0$; therefore the voltage error is not zero but is exactly the field voltage divided by the gain:

  $$E_{fd0} = K_A(V_{ref} - |V_0|), \qquad V_{ref} = |V_0| + \frac{E_{fd0}}{K_A}. \tag{16}$$

  Linearising (15) with $\Delta V_{ref} = 0$ gives

  $$\Delta E_{fd}(s) = -\frac{K_A}{1 + sT_A}\Delta|V|(s). \tag{17}$$

  Hence $\Delta|V| < 0$ produces $\Delta E_{fd} > 0$. The DC correction gain is 200 and the AVR pole is $-1/T_A = -50$ s^{-1}, mathematically explaining its fast voltage-support action. This is the *configuration published in the book*: Table 9.5 lists exactly 20 eigenvalues, matching the $4 \times 5 = 20$ state count. The inter-area swing mode is nevertheless very lightly damped. For $\lambda = \sigma + j\omega_d$,

  $$\zeta = -\frac{\sigma}{\sqrt{\sigma^2 + \omega_d^2}}. \tag{18}$$

  The computed AVR pole $-0.0042 + j4.4565$ therefore gives

  $$\zeta_{AVR} = \frac{0.0042}{\sqrt{0.0042^2 + 4.4565^2}} = 9.5 \times 10^{-4} = 0.095\%. \tag{19}$$

  The lightly damped conclusion thus follows from the computed pole.

- **Manual excitation/no AVR (4 states/machine, 16 total).** Removing the AVR does not remove the generator field circuit. Instead, the operator-supplied field voltage is frozen at the equilibrium value:

$$E_{fd}(t) = E_{fd0}, \qquad \Delta E_{fd}(s) = 0. \tag{20}$$

The manual state and input orders are therefore

$$x_i^{manual} = \begin{bmatrix}\delta_i & \omega_i & E'_{qi} & E'_{di}\end{bmatrix}^T, \qquad u_i^{manual} = \begin{bmatrix}P_{mi} & E_{fd0,i} & \rho\end{bmatrix}^T. \tag{21}$$

The four retained differential equations are

$$\dot{\delta}_i = \omega_B(\omega_i - 1), \tag{22}$$

$$\dot{\omega}_i = \frac{P_{mi} - P_{ei} - D_i(\omega_i - 1)}{2H_i}, \tag{23}$$

$$\dot{E}'_{qi} = \frac{E_{fd0,i} - E'_{qi} - (X_{di} - X'_{di})I_{di}}{T'_{d0i}}, \tag{24}$$

$$\dot{E}'_{di} = \frac{-E'_{di} + (X_{qi} - X'_{qi})I_{qi}}{T'_{q0i}}. \tag{25}$$

Thus manual excitation still has rotor, speed, d-axis flux, q-axis flux, stator-current, and network-voltage dynamics. What disappears is only the closed-loop field-voltage state and its terminal-voltage feedback.

Linearising the third equation in (25) gives

$$\Delta\dot{E}'_q = -\frac{1}{T'_{d0}}\Delta E'_q - \frac{X_d - X'_d}{T'_{d0}}\Delta I_d + \frac{1}{T'_{d0}}\underbrace{\Delta E_{fd}}_{0}. \tag{26}$$

Compare this with the AVR perturbation obtained from (15):

$$\Delta\dot{E}_{fd} = -\frac{1}{T_A}\Delta E_{fd} - \frac{K_A}{T_A}\Delta|V| + \frac{K_A}{T_A}\Delta V_{ref}. \tag{27}$$

With $\Delta V_{ref} = 0$, the AVR introduces the coupling $-(K_A/T_A)\Delta|V|$ and one additional state. Manual excitation has neither term. In state-matrix form this means

$$\begin{aligned}\Delta\dot{x}_{manual} &= A_{manual}\Delta x_{manual}, \quad A_{manual} \in \mathbb{R}^{16\times16},\\ \Delta\dot{x}_{AVR} &= A_{AVR}\Delta x_{AVR}, \qquad\quad A_{AVR} \in \mathbb{R}^{20\times20}.\end{aligned} \tag{28}$$

No-AVR also does not mean constant terminal voltage: V remains an algebraic solution of $g(x, y, u) = 0$ and changes whenever machine current or network topology changes. Only E_{fd} is constant.

This manual case is a separate 16-state study; it does *not* reproduce the 20-state Table 9.5. Its computed inter-area pole $-0.1173 + j4.0712$ gives

$$\zeta_{manual} = \frac{0.1173}{\sqrt{0.1173^2 + 4.0712^2}} = 0.0288 = 2.88\%. \tag{29}$$

The comparison is consequently equation-based: the AVR improves voltage regulation through (17), while adding the feedback block (27) changes the full system matrix and moves this inter-area pole from $\zeta = 2.88\%$ to $\zeta = 0.095\%$. This observation applies to the stated operating point and printed AVR parameters; it is not a universal claim that every AVR reduces damping.

The book uses the AVR throughout Section 9.6.1: Table 9.4 supplies the AVR parameters and Table 9.5 reports the 20-state eigenvalue spectrum. The manual case is a project-defined comparison, not a book configuration.

6 Small-signal stability

The complete DAE is linearised numerically around the in-house power-flow operating point. Algebraic network-voltage perturbations are eliminated with a Schur complement. The published eigenvalues are transcribed in the table below (Source data); they are not copied into the computed column and are not used to tune parameters.

6.1 Linearisation and Schur reduction

The nonlinear DAE in (5) is linearised about (x_0, y_0, u_0) after verifying

$$\|f(x_0, y_0, u_0)\|_\infty \le \varepsilon_{eq}, \qquad \|g(x_0, y_0, u_0)\|_\infty \le \varepsilon_{eq}. \tag{30}$$

SSSA fixes the external input, $\Delta u = 0$. Substituting $x = x_0 + \Delta x$ and $y = y_0 + \Delta y$ and retaining first-order terms gives

$$\begin{bmatrix} \Delta\dot{x} \\ 0 \end{bmatrix} = \begin{bmatrix} J_{xx} & J_{xy} \\ J_{yx} & J_{yy} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}. \tag{31}$$

The Jacobian blocks are

$$J_{xx} = \tfrac{\partial f}{\partial x}, \quad J_{xy} = \tfrac{\partial f}{\partial y}, \quad J_{yx} = \tfrac{\partial g}{\partial x}, \quad J_{yy} = \tfrac{\partial g}{\partial y}. \tag{32}$$

They are evaluated by central differences. For example,

$$J_{xx}(:, k) \approx \frac{f(x_0 + he_k, y_0, u_0) - f(x_0 - he_k, y_0, u_0)}{2h}, \tag{33}$$

$$J_{yx}(:, k) \approx \frac{g(x_0 + he_k, y_0, u_0) - g(x_0 - he_k, y_0, u_0)}{2h}, \tag{34}$$

with $h = 10^{-6}$; J_{xy} and J_{yy} use $y_0 \pm he_k$. These are derivatives of the DAE equations, not derivatives of the reference table.

The algebraic row is solved as

$$J_{yy}\Delta y = -J_{yx}\Delta x. \tag{35}$$

The code computes $K = J_{yy} \backslash J_{yx}$, avoiding an explicit inverse, and substitutes $\Delta y = -K\Delta x$ into the differential row:

$$A = J_{xx} - J_{xy}K = J_{xx} - J_{xy}J_{yy}^{-1}J_{yx}, \qquad \Delta\dot{x} = A\Delta x. \tag{36}$$

6.2 Eigenvalue solution and modal quantities

The solver then evaluates

$$Av_i = \lambda_i v_i, \qquad \det(\lambda_i I - A) = 0, qquad \lambda_i = \sigma_i + j\omega_{d,i}. \tag{37}$$

For each conjugate pair,

$$f_i = \frac{|\omega_{d,i}|}{2\pi}, \qquad \zeta_i = -\frac{\sigma_i}{\sqrt{\sigma_i^2 + \omega_{d,i}^2}}, \qquad \tau_i = -\frac{1}{\sigma_i} \quad (\sigma_i < 0). \tag{38}$$

A physical mode is asymptotically stable when $\sigma_i < 0$. Table 9 uses the full 20-state AVR matrix, matching the original-matrix column of Padiyar Table 9.5. The optional COI reduction is explicitly disabled in this execution, so the rotational reference-angle mode remains near the origin and is not misclassified as an unstable physical mode. Table 10 selects the positive-imaginary roots with $1 < \omega_d < 10$ rad/s; the omitted conjugates have identical f and ζ.

Reference matching occurs only after (36) has been solved. One-to-one assignment minimizes $|\lambda_{ours} - \lambda_{book}|$ without changing A or a model parameter. The reported errors are

$$\varepsilon_\lambda = 100\frac{|\lambda_{ours} - \lambda_{book}|}{|\lambda_{book}|}, \quad \varepsilon_f = 100\frac{|f_{ours} - f_{book}|}{f_{book}}, \quad \varepsilon_\zeta = 100\frac{|\zeta_{ours} - \zeta_{book}|}{|\zeta_{book}|}. \tag{39}$$

The eigenvalue percentage is omitted for the near-zero reference pair because its denominator makes a relative error numerically meaningless.

Table 9: Eigenvalue comparison: book values (λ_{book}, Padiyar Table 9.5) versus in-house computed values (λ_{ours}), with absolute difference $|\Delta\lambda|$ and swing-mode labels. Percentage error is omitted for near-zero reference-angle modes because it is not numerically meaningful. All 18 non-reference modes are stable.

No.	λ_{book} (s^{-1})	λ_{ours} (s^{-1})	$\lvert\Delta\lambda\rvert$ (s^{-1})	ε_λ (%)	Mode
1	-39.9893	-40.0225	0.0332	0.08	
2	-39.4922	-39.5270	0.0348	0.09	
3	$-24.5058 + 20.6749j$	$-24.5061 + 20.6319j$	0.0430	0.13	
4	$-24.5058 - 20.6749j$	$-24.5061 - 20.6319j$	0.0430	0.13	
5	$-25.0383 + 11.9973j$	$-25.0384 + 11.9598j$	0.0375	0.13	
6	$-25.0383 - 11.9973j$	$-25.0384 - 11.9598j$	0.0375	0.13	
7	-11.5835	-11.5566	0.0269	0.23	
8	-11.1735	-11.1485	0.0250	0.22	
9	$-0.7594 + 7.2938j$	$-0.7553 + 7.3157j$	0.0223	0.30	Swing 1
10	$-0.7594 - 7.2938j$	$-0.7553 - 7.3157j$	0.0223	0.30	Swing 1
11	$-0.7365 + 6.6899j$	$-0.7336 + 6.7108j$	0.0211	0.31	Swing 2
12	$-0.7365 - 6.6899j$	$-0.7336 - 6.7108j$	0.0211	0.31	Swing 2
13	$-0.0044 + 4.4444j$	$-0.0042 + 4.4565j$	0.0121	0.27	Inter-area
14	$-0.0044 - 4.4444j$	$-0.0042 - 4.4565j$	0.0121	0.27	Inter-area
15	-4.5727	-4.5701	0.0026	0.06	
16	-4.4802	-4.4776	0.0026	0.06	
17	-4.1121	-4.1095	0.0026	0.06	
18	$0.0000 + 0.0017j$	$0.0000 + 0.0000j$	0.0017	–	
19	$0.0000 - 0.0017j$	$0.0000 - 0.0000j$	0.0017	–	
20	-4.2449	-4.2423	0.0026	0.06	

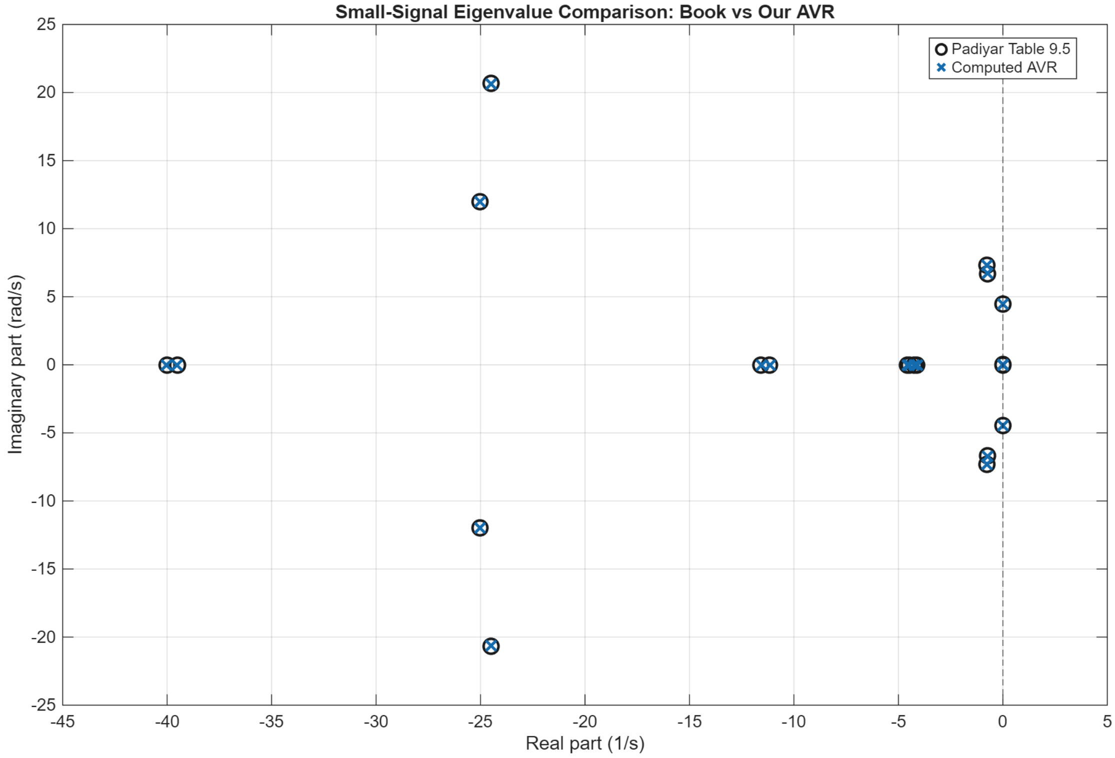

Figure 4: Graphical counterpart of Table 9 using only the AVR comparison on one complex-plane axis. Independently transcribed Padiyar Table 9.5 values are circles and computed AVR eigenvalues are crosses; coincident markers show direct graphical agreement.

Table 10: SSSA swing-mode comparison. Each AVR row directly compares the published and in-house eigenvalue, frequency, and damping ratio. Manual excitation has no published counterpart because it has a different state count; unavailable reference cells are shown with dashes. Both frequency and damping-ratio percentage errors are calculated directly from the displayed values.

Configuration	Mode	λ_{book} (s^{-1})	λ_{ours} (s^{-1})	f_{book} (Hz)	f_{ours} (Hz)	ε_f (%)	ζ_{book} (%)	ζ_{ours} (%)	ε_ζ (%)
AVR	Inter-area	$-0.0044 + 4.4444j$	$-0.0042 + 4.4565j$	0.7073	0.7093	0.27	0.099	0.095	3.79
AVR	Local Area 2	$-0.7365 + 6.6899j$	$-0.7336 + 6.7108j$	1.0647	1.0680	0.31	10.943	10.867	0.70
AVR	Local Area 1	$-0.7594 + 7.2938j$	$-0.7553 + 7.3157j$	1.1608	1.1643	0.30	10.356	10.269	0.83
Manual	Inter-area	–	$-0.1173 + 4.0712j$	–	0.6480	–	–	2.880	–
Manual	Local Area 2	–	$-0.4296 + 6.6775j$	–	1.0627	–	–	6.420	–
Manual	Local Area 1	–	$-0.4460 + 7.2336j$	–	1.1513	–	–	6.154	–

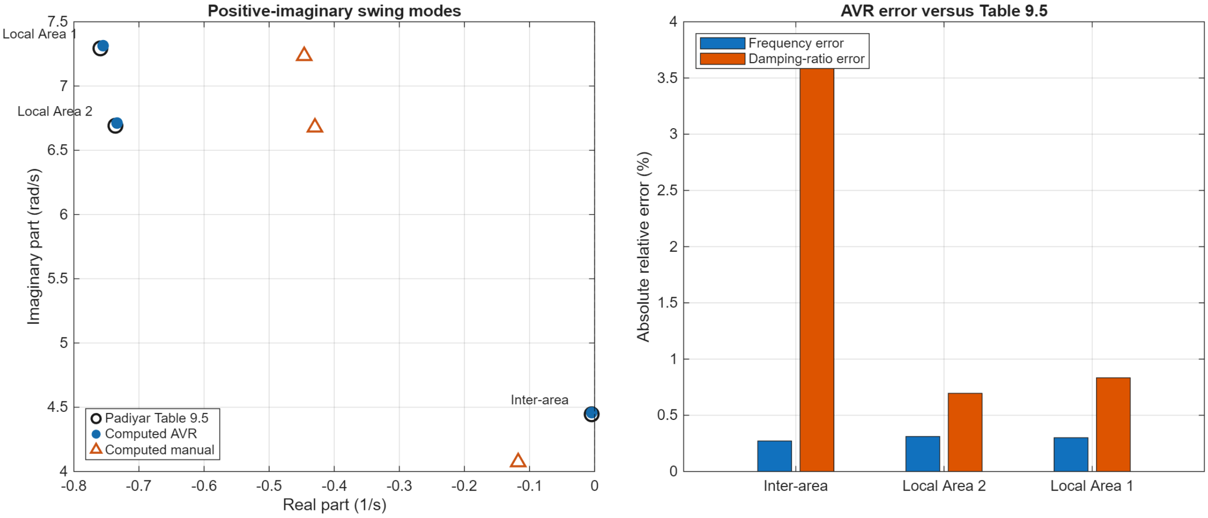

Figure 5: Graphical counterpart of Table 10. Left: positive-imaginary swing-mode eigenvalues. Right: AVR frequency and damping-ratio errors relative to the published values. The manual case is plotted only in the complex plane because no published manual-excitation target exists.

The AVR case reproduces the published mode groups without copying the table. Differences are reported as model/numerical discrepancies rather than removed by calibration. Manual excitation is a separate 16-state study and is not expected to reproduce the 20-state AVR table.

7 Time-Domain Simulation

The source pages specify no time-domain fault scenario. For a controlled comparison, this report defines a project scenario: a temporary balanced fault at bus 101, $Z_f = j(10 \times 10^{-5}) = j10^{-4}$ pu, applied at 1.0 s and cleared at 1.1 s, simulated from 0 to 20.0 s with $\Delta t = 0.005$ s. The same network, operating point and disturbance are used for manual and AVR configurations. The method uses a fixed nominal time step, exact event samples, and an iterative implicit-trapezoidal corrector.

7.1 Implicit trapezoidal integration

7.1.1 Why the implicit trapezoidal method is used

This choice is numerical, not a post-result fit to a desired trajectory. The implemented Padiyar model is a semi-explicit differential–algebraic system

$$\dot{x} = f(x, y, u), \qquad 0 = g(x, y, Y), \tag{40}$$

in which the machine states x evolve together with the algebraic network voltages y. Applying trapezoidal quadrature to the differential equation over one interval of length h gives

$$x_{n+1} = x_n + \frac{h}{2}\left[f(x_n, y_n, u_n) + f(x_{n+1}, y_{n+1}, u_{n+1})\right], \qquad g(x_{n+1}, y_{n+1}, Y_{n+1}) = 0. \tag{41}$$

Equation (41) is the equation actually solved by `stability.ts_step_kernel`; the algebraic endpoint condition is solved by `stability.ts_algebraic_solve`. The following rationale is a project derivation from that implemented equation, rather than a claim that a reference prints the project code verbatim.

First, for a smooth exact solution, Taylor expansion about t_n gives the one-step defect

$$x(t_n + h) - x(t_n) - \frac{h}{2}\left[\dot{x}(t_n) + \dot{x}(t_n + h)\right] = -\frac{h^3}{12}x^{(3)}(t_n) + \mathcal{O}(h^4). \tag{42}$$

Thus trapezoidal quadrature has one-step truncation error $\mathcal{O}(h^3)$ and global order two. It therefore improves the time-discretisation accuracy over first-order backward Euler at the same nominal step while retaining an implicit endpoint solve. Second, its linear stability can be seen without relying on a plot. For the test equation $\dot{x} = \lambda x$, with $z = h\lambda$, the update becomes

$$x_{n+1} = R(z)x_n, \qquad R(z) = \frac{1 + z/2}{1 - z/2}. \tag{43}$$

If $\operatorname{Re}(z) \le 0$, then $|1 - z/2|^2 - |1 + z/2|^2 = -2\operatorname{Re}(z) \ge 0$, so $|R(z)| \le 1$. The method is therefore A-stable: stable continuous modes are not made unstable merely because an explicit method's stability region is too small. This is useful for the coupled machine–network DAE, whose electrical and electromechanical modes can have substantially different time scales. The step size is nevertheless selected for accuracy and event resolution, not made arbitrarily large merely because the method is A-stable.

Third, the implicit endpoint formulation permits the network constraint $g = 0$ to be re-solved at each accepted endpoint. When a fault is applied or cleared, the integration grid lands exactly on the declared event time, the network topology is switched, and the algebraic voltage is re-solved under the new topology. Thus the stored event sample is consistent with the DAE rather than being obtained by interpolating across a discontinuous network change.

The method also has limitations. It is not L-stable because $R(z) \to -1$ as $z \to -\infty$; very fast decaying components are not annihilated in one large step and can appear as numerical ringing if h is chosen too large. Every step also requires nonlinear corrector and algebraic Newton iterations. The implementation therefore checks both the trapezoidal residual and the algebraic residual and fails closed when those iterations do not converge. These accuracy and convergence checks, rather than A-stability alone, justify an accepted TDS result.

The topology input is piecewise constant:

$$Y(t) = \begin{cases} Y_{pre}, & t < t_f, \\ Y_{fault}, & t_f \le t < t_c, \\ Y_{post}, & t \ge t_c, \end{cases} \qquad Y_{fault} = Y_{pre} + \frac{1}{Z_f}e_f e_f^T. \tag{44}$$

Here e_f selects the faulted bus. Times t_f and t_c are inserted exactly into the nominal Δt grid, so no step hides a topology transition.

At each accepted state x, the network voltage is obtained from

$$g(x, y, Y) = 0. \tag{45}$$

For algebraic Newton iteration k, set $r_g^{(k)} = g(x, y^{(k)}, Y)$ and $G_y^{(k)} = \partial g/\partial y$. Then

$$G_y^{(k)}\Delta y^{(k)} = -r_g^{(k)}, \qquad y^{(k+1)} = y^{(k)} + \alpha_k \Delta y^{(k)}. \tag{46}$$

The line search tries $\alpha_k = 1, 1/2, 1/4, \ldots, 2^{-16}$ and accepts the first value that decreases $\|r_g\|_\infty$. The algebraic solve requires $\|r_g\|_\infty \le 10^{-11}$ and refreshes G_y by central differences if a step is rejected.

For $h = t_{n+1} - t_n$ and $f_n = f(x_n, y_n, u_n)$, the predictor is

$$\hat{x}_{n+1}^{(0)} = x_n + hf_n, \qquad \hat{y}_{n+1}^{(0)} = y_n. \tag{47}$$

Corrector iteration c performs the following sequence:

$$g(\hat{x}_{n+1}^{(c)}, \hat{y}_{n+1}^{(c)}, Y_n) = 0, \tag{48}$$

$$x_{n+1}^{(c+1)} = x_n + \frac{h}{2}\left[f_n + f(\hat{x}_{n+1}^{(c)}, \hat{y}_{n+1}^{(c)}, u_n)\right], \tag{49}$$

$$g(x_{n+1}^{(c+1)}, y_{n+1}^{(c+1)}, Y_n) = 0, \tag{50}$$

$$R_{trap}^{(c+1)} = x_{n+1}^{(c+1)} - x_n - \frac{h}{2}\left[f_n + f(x_{n+1}^{(c+1)}, y_{n+1}^{(c+1)}, u_n)\right]. \tag{51}$$

The next iterate uses $\hat{x}_{n+1}^{(c+1)} = x_{n+1}^{(c+1)}$. Acceptance requires both

$$\|x_{n+1}^{(c+1)} - \hat{x}_{n+1}^{(c)}\|_\infty \le \tau_n, \qquad \|R_{trap}^{(c+1)}\|_\infty \le \tau_n, \tag{52}$$

where the implemented mixed tolerance is

$$\tau_n = 10^{-10} + 10^{-8} \max\left(1, \|x_{n+1}^{(c+1)}\|_\infty\right). \tag{53}$$

At most 12 corrector iterations are allowed. A failed corrector raises an error instead of accepting the step. After acceptance, $Y(t_{n+1})$ is selected again and (45) is solved, ensuring that a stored event sample is consistent with the new topology.

7.2 Fixed corrector iteration versus adaptive time stepping

The corrector iteration and the time-step controller solve different problems. For fixed h, repeated correction drives the nonlinear equation residual

$$F_h(x_{n+1}) = x_{n+1} - x_n - \frac{h}{2}(f_n + f_{n+1}) \tag{54}$$

toward zero. Thus $\|F_h\|_\infty \ll 1$ proves that the trapezoidal equation for the chosen h was solved accurately. It does not estimate the discretization error $\|x(t_{n+1}) - x_{n+1}\|$.

An adaptive second-order trapezoidal method additionally needs an independent local-error estimate. Step doubling would compute one full step x_h and two half steps $x_{h/2,h/2}$, then form

$$e_n = \frac{x_{h/2,h/2} - x_h}{2^2 - 1}, \qquad E_n = \left\|\frac{e_n}{a_{tol} + r_{tol}\max(|x_n|, |x_{h/2,h/2}|)}\right\|_\infty. \tag{55}$$

The step is accepted only when $E_n \le 1$; otherwise it is rejected and recomputed. A typical controller is

$$h_{new} = h \text{ clip}\left(f_{min}, f_{max}, safety\, E_n^{-1/3}\right), \tag{56}$$

with event times imposing an upper bound on h_{new} so the solver still lands exactly on t_f and t_c.

The present validation solver intentionally remains fixed-step because the published comparison protocol requires a deterministic common output grid and the implementation currently has neither the independent estimator (55) nor step rejection. Calling corrector iteration adaptive would therefore be false. Fixed $h = 0.005$ s is accepted here only after equilibrium, residual, convergence, and finite-trajectory checks; it is not presented as a general proof of time-discretization accuracy. Adaptive stepping is a valid future production improvement, but it must add the estimator, accept/reject controller, exact event landing, resampling for cross-tool comparison, and convergence tests under successively tighter tolerances before replacing this reproducible validation path.

The recorded outputs are reconstructed from the accepted (x_n, y_n):

$$|V_b| = \sqrt{y_{2b-1}^2 + y_{2b}^2}, \qquad P_{ei} = V_{di}I_{di} + V_{qi}I_{qi} + R_a(I_{di}^2 + I_{qi}^2), \tag{57}$$

with $P_{ei}[\mathrm{MW}] = S_{base}[\mathrm{MVA}]P_{ei}[\mathrm{pu}]$. The saved diagnostics include the initial DAE residual, corrector update norm, trapezoidal residual, iteration count, convergence flag, and non-converged-step count; consequently, a plausible plot cannot conceal a failed numerical step.

Table 11: Time-domain-simulation input contract, numerical status, and response extrema for the 0–20 s bus-101 fault scenario with $Z_f = j10^{-4}$ pu, $t_f = 1.0$ s, $t_c = 1.1$ s, and $\Delta t = 0.005$ s.

Metric	Unit	AVR	Manual excitation
Fault bus	bus ID	101	101
Fault impedance Z_f	pu	$0 + 0.0001j$	$0 + 0.0001j$
Simulation horizon	s	20.000	20.000
Fixed time step	s	0.0050	0.0050
Initial DAE residual	pu	9.215e-13	5.684e-14
Non-converged steps	steps	0	0
Maximum corrector residual	pu	1.301e-07	5.438e-08
Minimum bus voltage	pu	0.0036	0.0033
Maximum speed deviation	pu	7.676e-03	1.710e-02

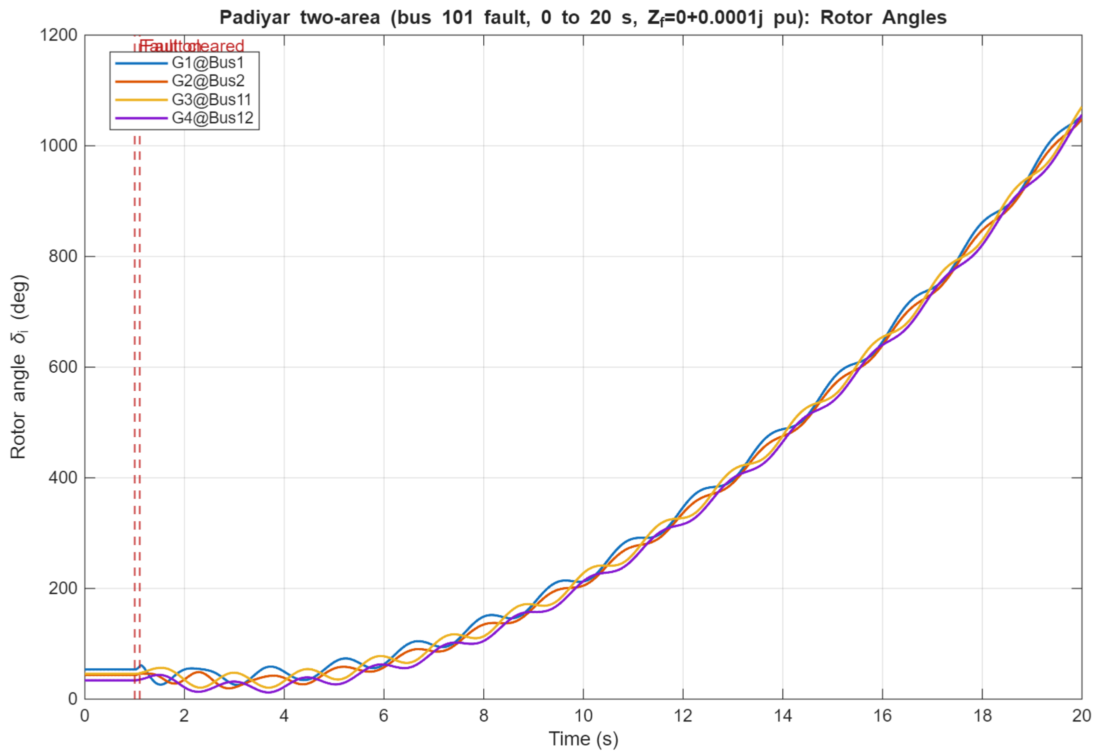

Figure 6: Bus-101 project fault response using the AVR configuration (the book's 20-state model): stored absolute generator rotor angles $\delta_i(t)$ in degrees. No initial-angle subtraction is applied. Red markers identify fault application and clearance. The simulated interval is 0–20 s and $Z_f = j10^{-4}$ pu.

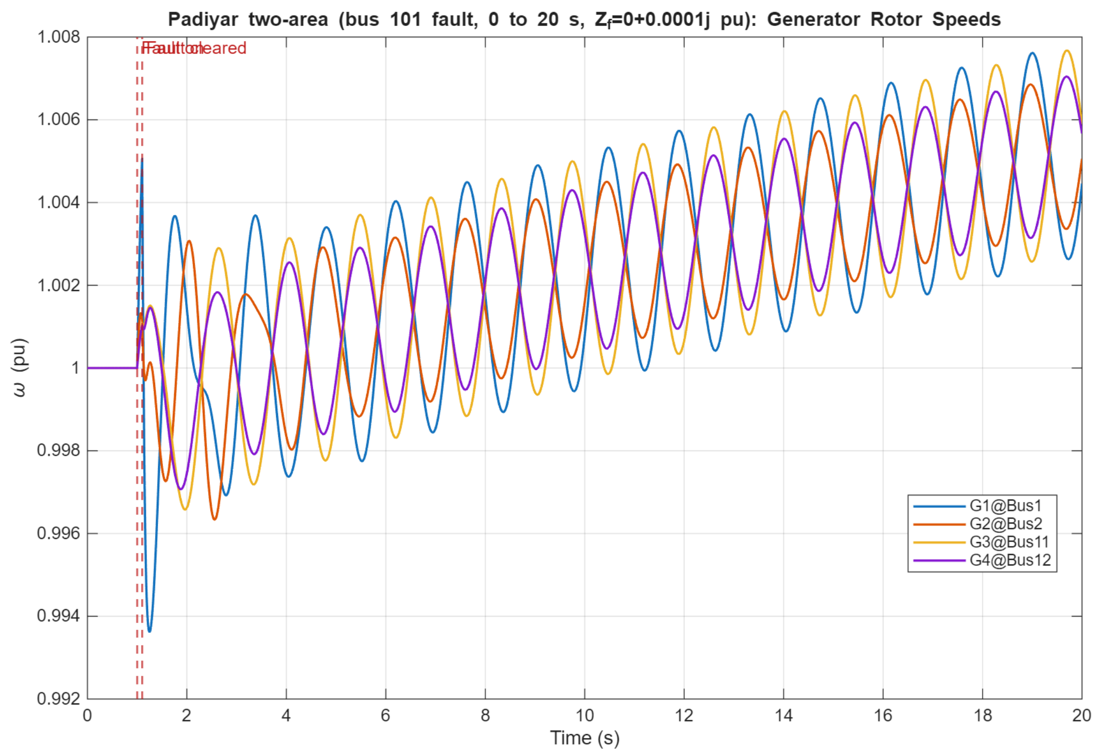

Figure 7: Bus-101 project fault response using the AVR configuration: stored absolute rotor speeds $\omega_i(t)$ in per unit. The plotted values are the TS result values themselves; the graph does not plot $\omega_i - 1$. The simulated interval is 0–20 s and $Z_f = j10^{-4}$ pu.

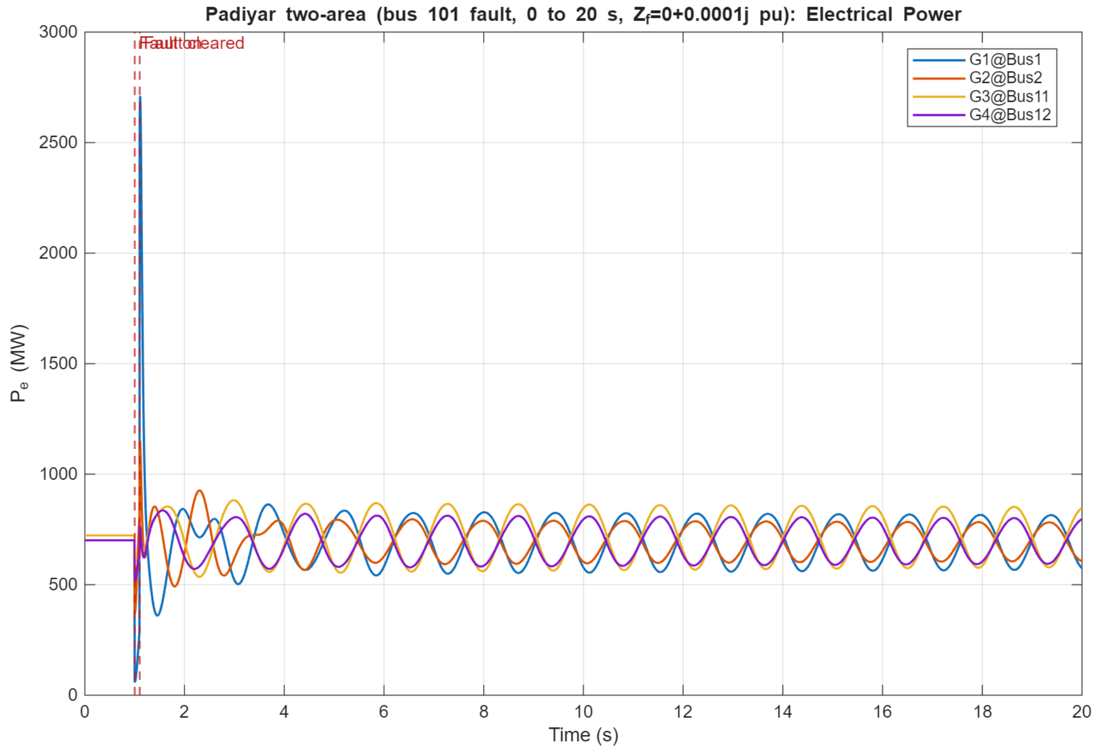

Figure 8: Bus-101 project fault response using the AVR configuration: generator air-gap electrical powers $P_{ei}(t)$ reconstructed from the accepted TS states and algebraic bus voltages. The simulated interval is 0–20 s and $Z_f = j10^{-4}$ pu.

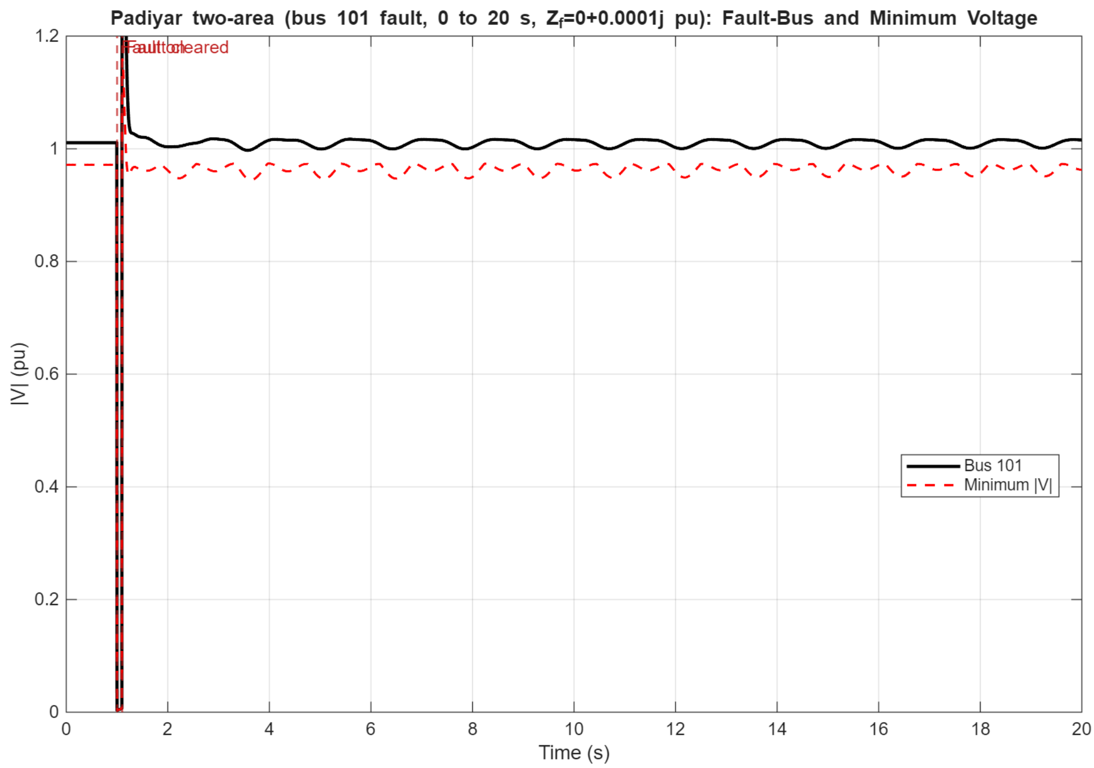

Figure 9: Bus-101 project fault response using the AVR configuration: fault-bus voltage magnitude and the minimum voltage magnitude across the network. The simulated interval is 0–20 s and $Z_f = j10^{-4}$ pu.

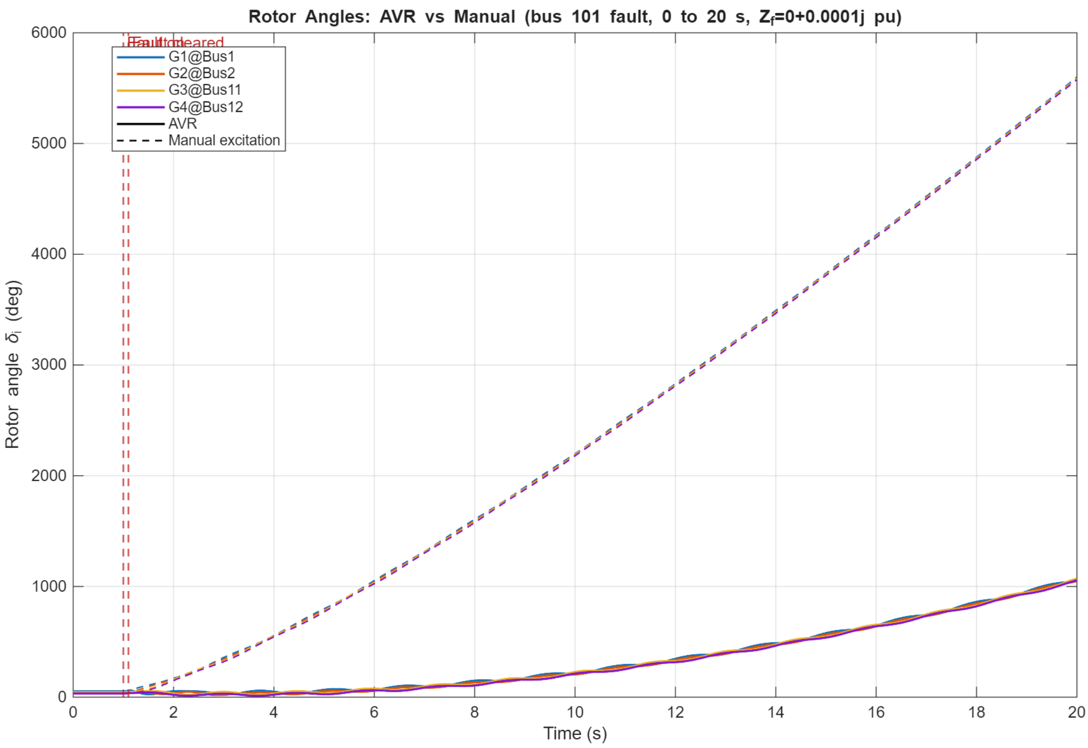

Figure 10: Stored absolute rotor angles for the bus-101 fault: AVR (solid) versus manual excitation (dashed). Each colour identifies one generator. No initial-angle subtraction is applied. The simulated interval is 0–20 s and $Z_f = j10^{-4}$ pu.

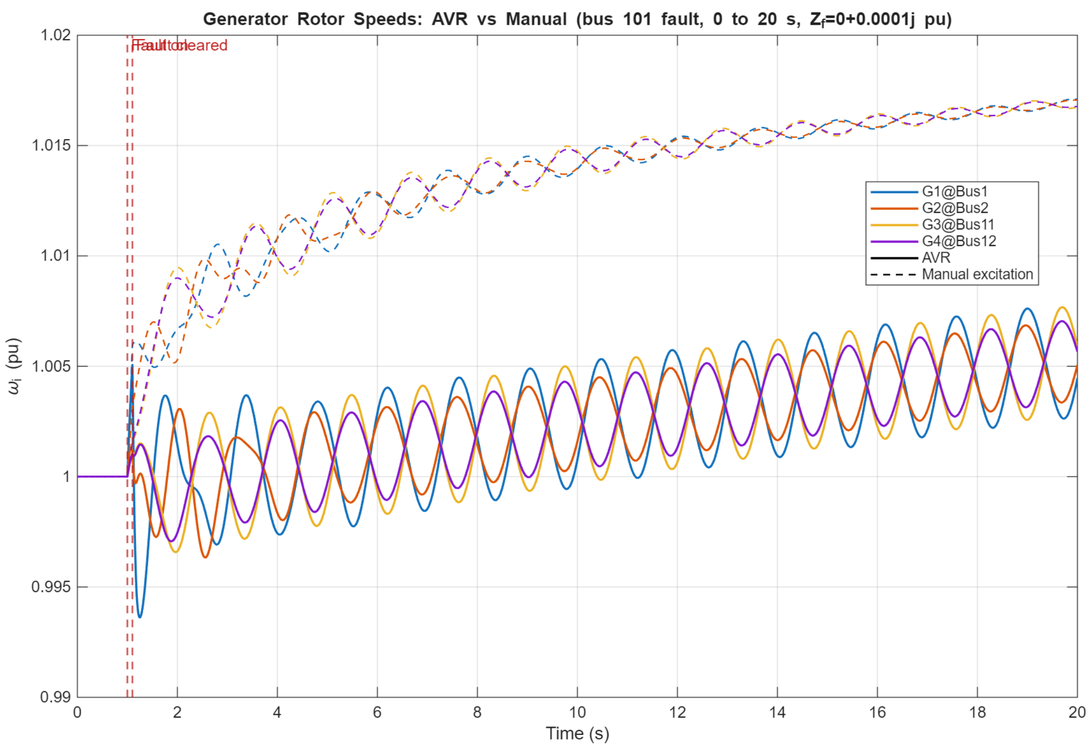

Figure 11: Stored absolute rotor speeds $\omega_i(t)$ for the bus-101 fault: AVR (solid) versus manual excitation (dashed). No $\omega_i - 1$ plotting transform is applied. The simulated interval is 0–20 s and $Z_f = j10^{-4}$ pu.

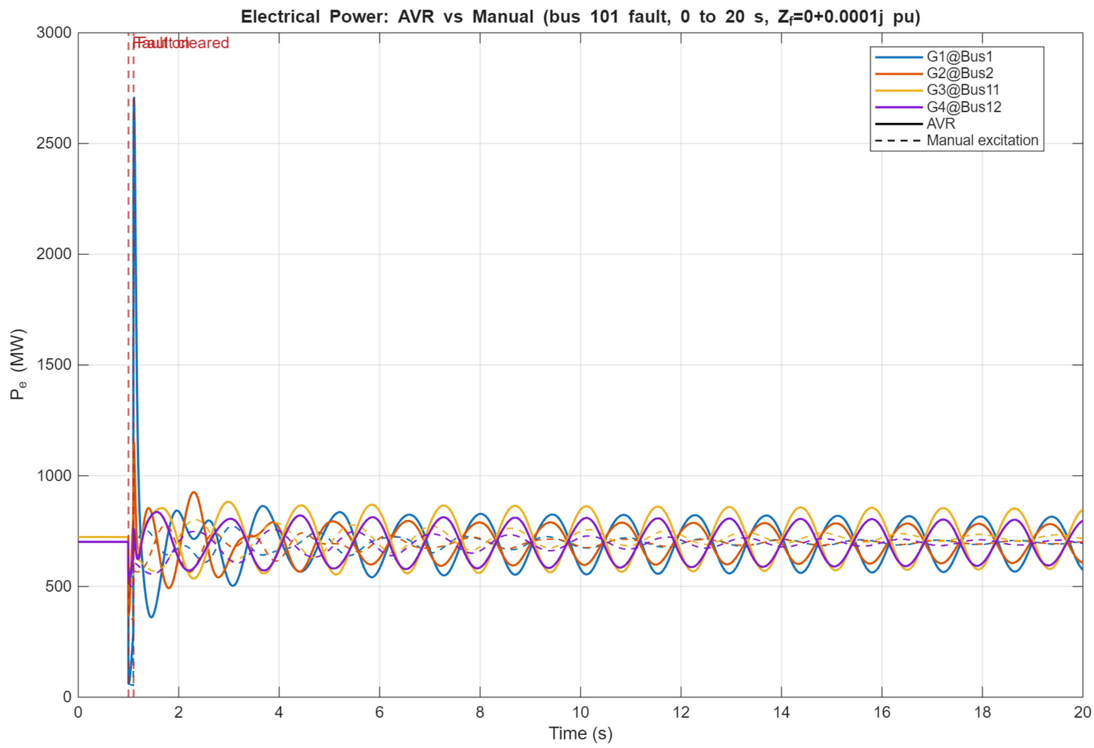

Figure 12: Generator electrical powers for the bus-101 fault: AVR (solid) versus manual excitation (dashed). The simulated interval is 0–20 s and $Z_f = j10^{-4}$ pu.

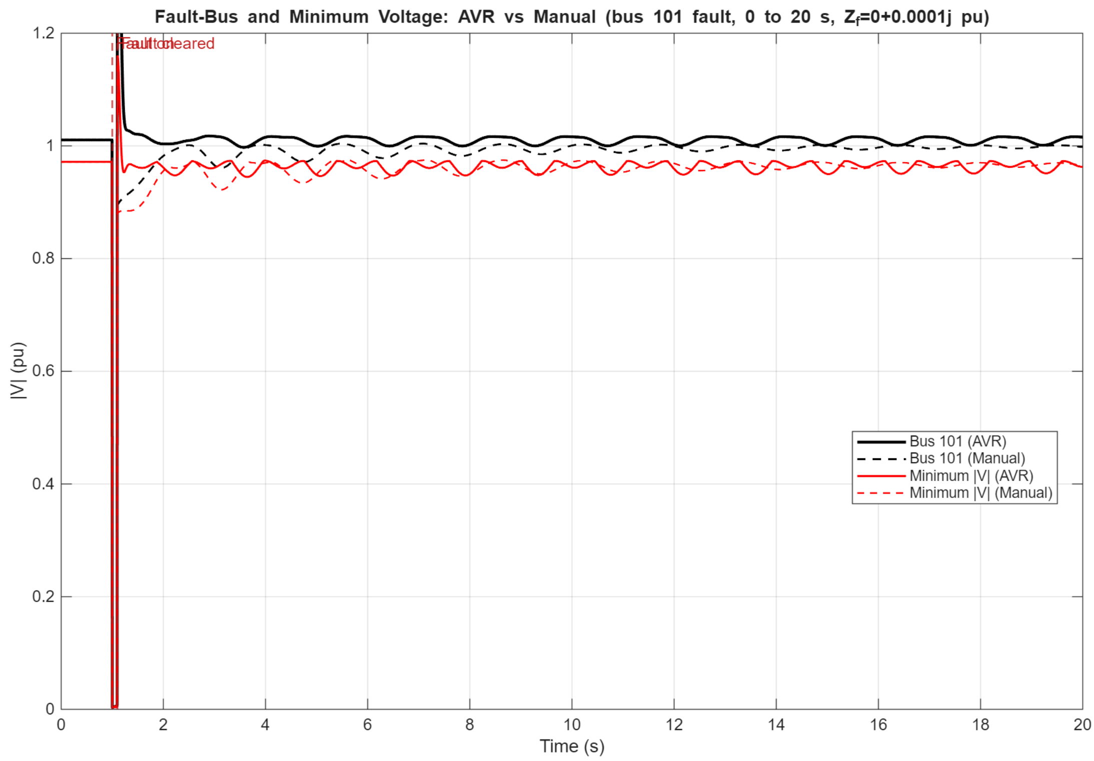

Figure 13: Fault-bus and minimum network voltage magnitudes for the bus-101 fault: AVR (solid) versus manual excitation (dashed). Quantitative modal damping is determined from (38) and Table 10, not inferred from visual appearance. The simulated interval is 0–20 s and $Z_f = j10^{-4}$ pu.

8 IEEE 14-bus cross-validation against PSAT

This section cross-validates the common classical PF, SSSA and TDS substrate against PSAT 2.1.11. It is an independent infrastructure check and is not a claim that PSAT implements the five-state Padiyar model used in the preceding sections. Every IEEE14 table and figure below is generated in the same invocation as a fresh in-house run and a fresh PSAT run; no saved trajectory, prior numeric table, or prior plot is read. The generating entry point is `generate_padiyar_ieee14_psat_tables`.

8.1 System description

The network is the MATPOWER 6 IEEE 14-bus case converted into the project schema. It has 14 bus rows, five in-service generator rows at buses 1, 2, 3, 6 and 8, and 20 in-service branch rows. The complete input inventory used in this study is given before the computed PF, SSSA and TDS results.

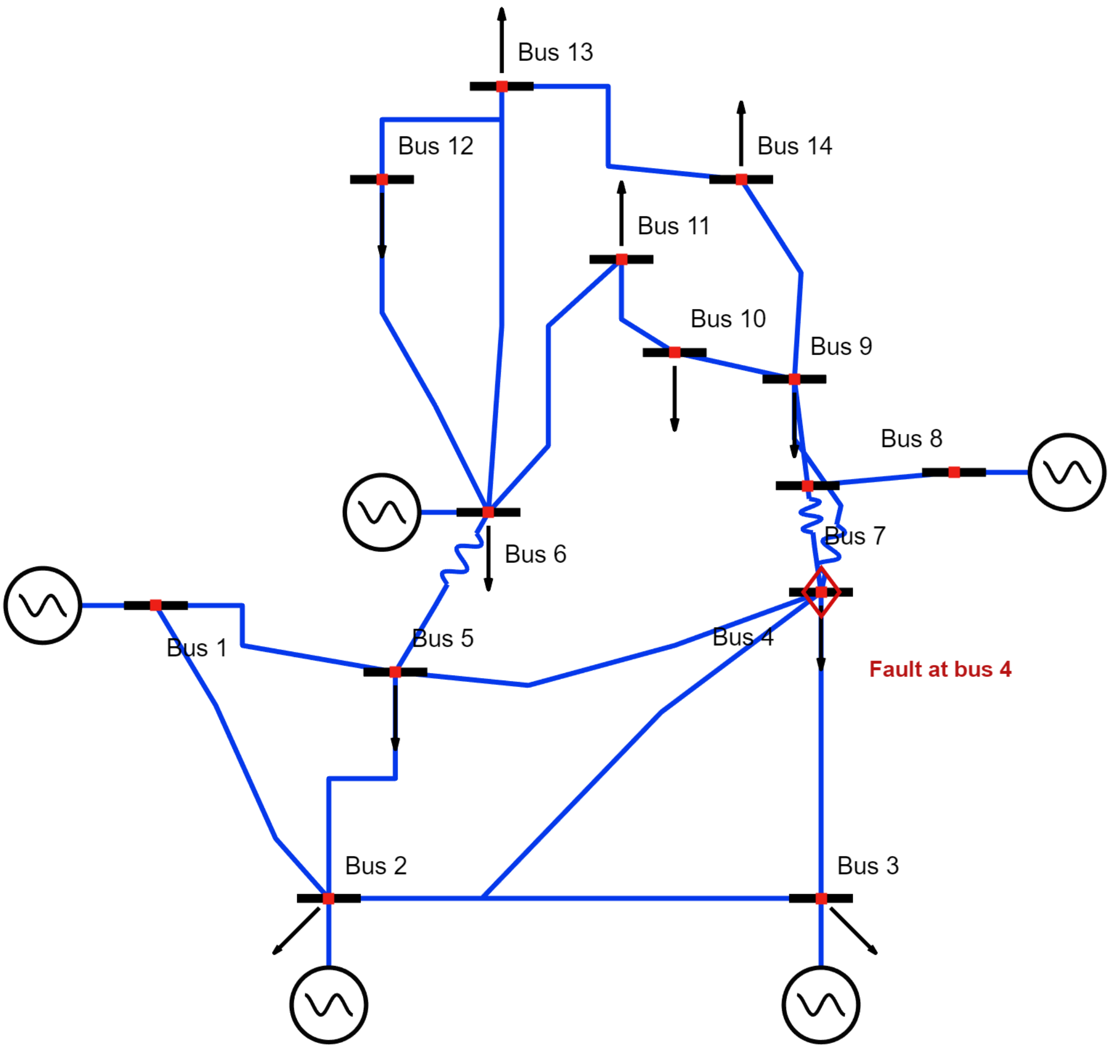

Figure 14: IEEE 14-bus network generated from the same MATPOWER branch, bus and generator rows used by Our and PSAT. Generator buses, nonzero-load buses, transformer branches and the bus-4 fault location are marked explicitly.

8.2 Source data (MATPOWER IEEE 14-bus)

Table 12: MATPOWER IEEE14 case summary: 14 buses, five online generators and 20 line/transformer rows, with the shared base quantities.

Item	Value	Contract note
Case	MATPOWER 6.0 IEEE 14-bus	Converted from the supplied case14.m
Power base	100 MVA	MATPOWER mpc.baseMVA
Nominal frequency	60 Hz	Project dynamic base
Bus rows	14	1 REF, 4 PV, 9 PQ
Online generators	5	Buses [1,2,3,6,8]
Online branches	20	3 transformer-tap branches
Total specified demand	259.0 MW, 73.5 MVAr	Sum of bus Pd and Qd
Nonzero shunt susceptance	19.0 MVAr	Bus 9 source Bs entry

8.2.1 Bus data and bus types

Table 14 records the source bus operating entries and aggregated source generation. Table 15 preserves the remaining MATPOWER bus-matrix fields, including the bus type, shunt, area, voltage limits and initial voltage. The source P_G/Q_G entries are case data; their input/output interpretation follows the effective REF, PV and PQ bus types.

Table 13: IEEE14 bus-type assignment, including the MATPOWER type codes and the buses in each REF, PV and PQ group.

Bus type	MATPOWER code	Bus IDs	Count
REF	3	[1]	1
PV	2	[2,3,6,8]	4
PQ	1	[4,5,7,9,10,11,12,13,14]	9

Table 14: IEEE 14-bus data and bus types shared by the in-house and PSAT runs.

Bus	Type	$\|V\|_{src}$ (pu)	θ_{src} (deg)	P_L (MW)	Q_L (MVAr)	P_G (MW)	Q_G (MVAr)
1	REF	1.0600	0.0000	0.00	0.00	232.40	-16.90
2	PV	1.0450	-4.9800	21.70	12.70	40.00	42.40
3	PV	1.0100	-12.7200	94.20	19.00	0.00	23.40
4	PQ	1.0190	-10.3300	47.80	-3.90	0.00	0.00
5	PQ	1.0200	-8.7800	7.60	1.60	0.00	0.00
6	PV	1.0700	-14.2200	11.20	7.50	0.00	12.20
7	PQ	1.0620	-13.3700	0.00	0.00	0.00	0.00
8	PV	1.0900	-13.3600	0.00	0.00	0.00	17.40
9	PQ	1.0560	-14.9400	29.50	16.60	0.00	0.00
10	PQ	1.0510	-15.1000	9.00	5.80	0.00	0.00
11	PQ	1.0570	-14.7900	3.50	1.80	0.00	0.00
12	PQ	1.0550	-15.0700	6.10	1.60	0.00	0.00
13	PQ	1.0500	-15.1600	13.50	5.80	0.00	0.00
14	PQ	1.0360	-16.0400	14.90	5.00	0.00	0.00

Table 15: MATPOWER IEEE14 bus data: complete bus-matrix parameters supplied to the case converter.

Bus	Type	P_d (MW)	Q_d (MVAr)	G_s (MW)	B_s (MVAr)	Area	V_m (pu)	V_a (deg)	baseKV (kV)	Zone	V_{max} (pu)	V_{min} (pu)
1	REF	0.00	0.00	0.00	0.00	1	1.0600	0.0000	0.0	1	1.06	0.94
2	PV	21.70	12.70	0.00	0.00	1	1.0450	-4.9800	0.0	1	1.06	0.94
3	PV	94.20	19.00	0.00	0.00	1	1.0100	-12.7200	0.0	1	1.06	0.94
4	PQ	47.80	-3.90	0.00	0.00	1	1.0190	-10.3300	0.0	1	1.06	0.94
5	PQ	7.60	1.60	0.00	0.00	1	1.0200	-8.7800	0.0	1	1.06	0.94
6	PV	11.20	7.50	0.00	0.00	1	1.0700	-14.2200	0.0	1	1.06	0.94
7	PQ	0.00	0.00	0.00	0.00	1	1.0620	-13.3700	0.0	1	1.06	0.94
8	PV	0.00	0.00	0.00	0.00	1	1.0900	-13.3600	0.0	1	1.06	0.94
9	PQ	29.50	16.60	0.00	19.00	1	1.0560	-14.9400	0.0	1	1.06	0.94
10	PQ	9.00	5.80	0.00	0.00	1	1.0510	-15.1000	0.0	1	1.06	0.94
11	PQ	3.50	1.80	0.00	0.00	1	1.0570	-14.7900	0.0	1	1.06	0.94
12	PQ	6.10	1.60	0.00	0.00	1	1.0550	-15.0700	0.0	1	1.06	0.94
13	PQ	13.50	5.80	0.00	0.00	1	1.0500	-15.1600	0.0	1	1.06	0.94
14	PQ	14.90	5.00	0.00	0.00	1	1.0360	-16.0400	0.0	1	1.06	0.94

8.2.2 Line data

All 20 in-service MATPOWER line/transformer rows are transcribed in Table 16. A zero source tap ratio denotes the MATPOWER unity-ratio default; the three nonzero ratios identify the transformer branches.

Table 16: MATPOWER IEEE14 line and transformer data (branch matrix); all 20 rows are in service.

From	To	r (pu)	x (pu)	b (pu)	rateA (MVA)	rateB (MVA)	rateC (MVA)	ratio (pu)	shift (deg)	Status	θ_{min} (deg)	θ_{max} (deg)
1	2	0.01938	0.05917	0.0528	0	0	0	0.000	0.0	1	-360	360
1	5	0.05403	0.22304	0.0492	0	0	0	0.000	0.0	1	-360	360
2	3	0.04699	0.19797	0.0438	0	0	0	0.000	0.0	1	-360	360
2	4	0.05811	0.17632	0.0340	0	0	0	0.000	0.0	1	-360	360
2	5	0.05695	0.17388	0.0346	0	0	0	0.000	0.0	1	-360	360
3	4	0.06701	0.17103	0.0128	0	0	0	0.000	0.0	1	-360	360
4	5	0.01335	0.04211	0.0000	0	0	0	0.000	0.0	1	-360	360
4	7	0.00000	0.20912	0.0000	0	0	0	0.978	0.0	1	-360	360
4	9	0.00000	0.55618	0.0000	0	0	0	0.969	0.0	1	-360	360
5	6	0.00000	0.25202	0.0000	0	0	0	0.932	0.0	1	-360	360
6	11	0.09498	0.19890	0.0000	0	0	0	0.000	0.0	1	-360	360
6	12	0.12291	0.25581	0.0000	0	0	0	0.000	0.0	1	-360	360
6	13	0.06615	0.13027	0.0000	0	0	0	0.000	0.0	1	-360	360
7	8	0.00000	0.17615	0.0000	0	0	0	0.000	0.0	1	-360	360
7	9	0.00000	0.11001	0.0000	0	0	0	0.000	0.0	1	-360	360
9	10	0.03181	0.08450	0.0000	0	0	0	0.000	0.0	1	-360	360
9	14	0.12711	0.27038	0.0000	0	0	0	0.000	0.0	1	-360	360
10	11	0.08205	0.19207	0.0000	0	0	0	0.000	0.0	1	-360	360
12	13	0.22092	0.19988	0.0000	0	0	0	0.000	0.0	1	-360	360
13	14	0.17093	0.34802	0.0000	0	0	0	0.000	0.0	1	-360	360

8.2.3 Generator data and dynamic parameters

The five online generator rows are transcribed in Table 17. Generator-cost rows are retained for case completeness in Table 18, although no optimal-power-flow calculation is performed here.

Table 17: MATPOWER IEEE14 online generator data.

Gen	Bus	P_g (MW)	Q_g (MVAr)	Q_{max} (MVAr)	Q_{min} (MVAr)	V_g (pu)	mBase (MVA)	Status	P_{max} (MW)	P_{min} (MW)
1	1	232.40	-16.90	10.00	0.00	1.0600	100.0	1	332.40	0.00
2	2	40.00	42.40	50.00	-40.00	1.0450	100.0	1	140.00	0.00
3	3	0.00	23.40	40.00	0.00	1.0100	100.0	1	100.00	0.00
4	6	0.00	12.20	24.00	-6.00	1.0700	100.0	1	100.00	0.00
5	8	0.00	17.40	24.00	-6.00	1.0900	100.0	1	100.00	0.00

Table 18: MATPOWER IEEE14 polynomial generator-cost data retained from the source case.

Gen	Bus	Model	Startup	Shutdown	N	c_2	c_1	c_0
1	1	2	0.0	0.0	3	0.0430292599	20	0
2	2	2	0.0	0.0	3	0.25	20	0
3	3	2	0.0	0.0	3	0.01	40	0
4	6	2	0.0	0.0	3	0.01	40	0
5	8	2	0.0	0.0	3	0.01	40	0

The static MATPOWER case does not supply a complete dynamic data set. Therefore the five classical-machine rows in Table 19 are explicitly labelled `ASSUMED_DIAGNOSTIC`; the identical rows are passed to Our and PSAT.

Table 19: Classical-machine parameters used identically by Our and PSAT.

Machine	Bus	H (s)	D (pu)	X'_d (pu)	Provenance
1	1	5.00	0.00	0.300	ASSUMED_DIAGNOSTIC
2	2	5.00	0.00	0.300	ASSUMED_DIAGNOSTIC
3	3	5.00	0.00	0.300	ASSUMED_DIAGNOSTIC
4	6	5.00	0.00	0.300	ASSUMED_DIAGNOSTIC
5	8	5.00	0.00	0.300	ASSUMED_DIAGNOSTIC

8.2.4 Fault and simulation inputs

The identical event contract in Table 20 uses a balanced three-phase fault at bus 4 with $Z_f = j10^{-4}$ pu, applied at 1.0 s and cleared at 1.1 s. Both simulations run to 15 s with a nominal 0.01 s step.

Table 20: Identical IEEE14 model and event inputs supplied to Our and PSAT.

Input	Our	PSAT
Network	MATPOWER 6 IEEE14	Converted from the same case
Generator buses	[1,2,3,6,8]	[1,2,3,6,8]
Classical dynamics (H, D, X'_d)	$(5, 0, 0.3)$	$(5, 0, 0.3)$
Fault bus	4	4
Fault impedance Z_f (pu)	$j10^{-4}$	$j10^{-4}$
Fault interval (s)	1.0–1.1	1.0–1.1
Simulation horizon (s)	15	15
Nominal step (s)	0.01	0.01

8.3 Power-flow results

Table 21 reports both solved bus-voltage columns directly, with bus IDs mapped before subtraction. PSAT receives the network through the project converter rather than through an unrelated built-in case.

Table 21: Fresh IEEE14 power-flow comparison: Our versus PSAT.

Bus	Our $\lvert V\rvert$ (pu)	Our θ (deg)	PSAT $\lvert V\rvert$ (pu)	PSAT θ (deg)	Absolute difference $\lvert\Delta V\rvert$ (pu)	Absolute difference $\lvert\Delta\theta\rvert$ (deg)
1	1.060000	0.000000	1.060000	0.000000	0.00e+00	0.00e+00
2	1.045000	-4.982589	1.045000	-4.982589	0.00e+00	2.66e-15
3	1.010000	-12.725100	1.010000	-12.725100	0.00e+00	1.78e-15
4	1.017671	-10.312901	1.017671	-10.312901	0.00e+00	1.78e-15
5	1.019514	-8.773854	1.019514	-8.773854	2.22e-16	5.33e-15
6	1.070000	-14.220946	1.070000	-14.220946	0.00e+00	1.95e-14
7	1.061520	-13.359627	1.061520	-13.359627	0.00e+00	1.78e-15
8	1.090000	-13.359627	1.090000	-13.359627	0.00e+00	3.55e-15
9	1.055932	-14.938521	1.055932	-14.938521	4.44e-16	8.88e-15
10	1.050985	-15.097288	1.050985	-15.097288	6.66e-16	1.24e-14
11	1.056907	-14.790622	1.056907	-14.790622	4.44e-16	1.95e-14
12	1.055189	-15.075585	1.055189	-15.075585	2.22e-16	2.84e-14
13	1.050382	-15.156276	1.050382	-15.156276	4.44e-16	2.84e-14
14	1.035530	-16.033645	1.035530	-16.033645	6.66e-16	1.42e-14

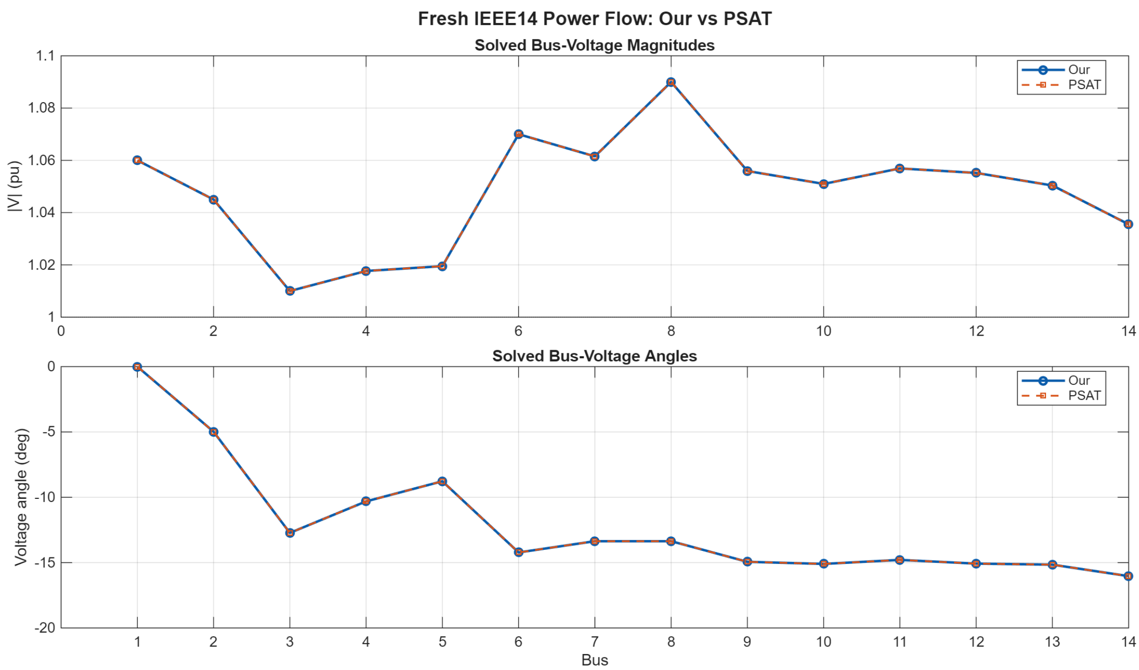

Figure 15: Fresh IEEE14 solved bus-voltage magnitudes and angles. Our is solid and PSAT is dashed; both traces use the bus-ID mapping reported in Table 21.

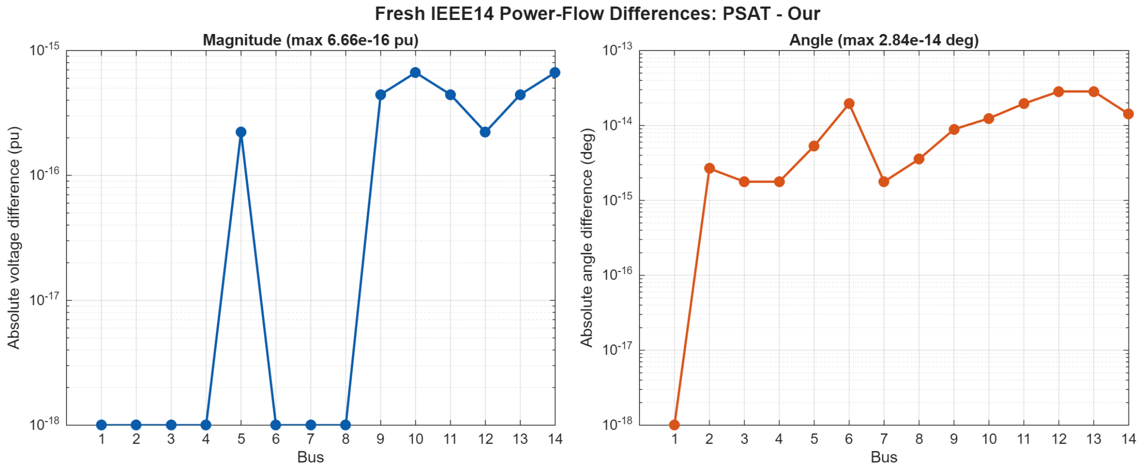

Figure 16: Per-bus absolute PF differences from the same fresh Our and PSAT solutions. Zero differences are shown at the numerical display floor on the logarithmic axes.

8.4 Small-signal stability results

Table 22 compares the positive-imaginary physical relative modes from the identical five-machine classical model. The common-angle zero mode is omitted by the declared 10^{-3} rad/s frequency threshold. PSAT values are reported exactly as returned by its state-matrix analysis; its `fm_eigen` implementation applies a documented -10^{-6} diagonal shift. No equation is repeated here because the model and linearisation have already been described above.

Table 22: Fresh IEEE14 SSSA modal comparison: Our versus raw PSAT output.

Mode	λ_{Our} (s^{-1})	λ_{PSAT} (s^{-1})	$\|\Delta\lambda\|$ (s^{-1})	f_{Our} (Hz)	f_{PSAT} (Hz)	ζ_{Our} (%)	ζ_{PSAT} (%)
1	$-0.000000 + 11.520207j$	$-0.000001 + 11.520207j$	1.000e-06	1.83350	1.83350	0.00000	0.00001
2	$0.000000 + 9.955055j$	$-0.000001 + 9.955055j$	1.000e-06	1.58440	1.58440	-0.00000	0.00001
3	$-0.000000 + 9.621911j$	$-0.000001 + 9.621911j$	1.000e-06	1.53137	1.53137	0.00000	0.00001
4	$0.000000 + 8.505808j$	$-0.000001 + 8.505808j$	1.000e-06	1.35374	1.35374	-0.00000	0.00001

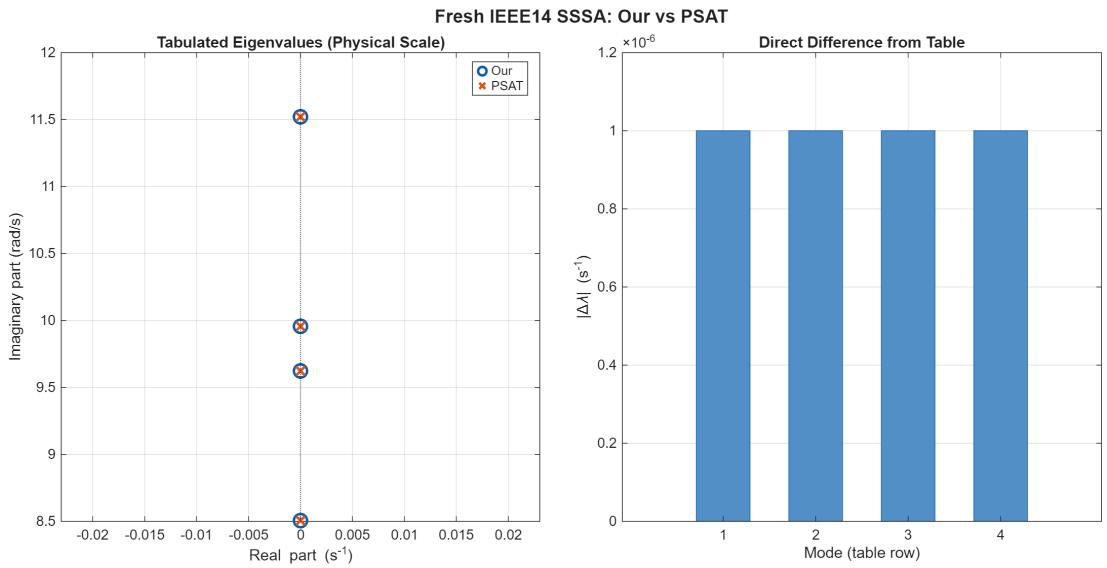

Figure 17: Fresh IEEE14 positive-imaginary modes plotted directly from Table 22. The complex-plane panel uses the unscaled eigenvalues in physical units; the difference panel reports the corresponding $|\Delta\lambda|$ for each table row.

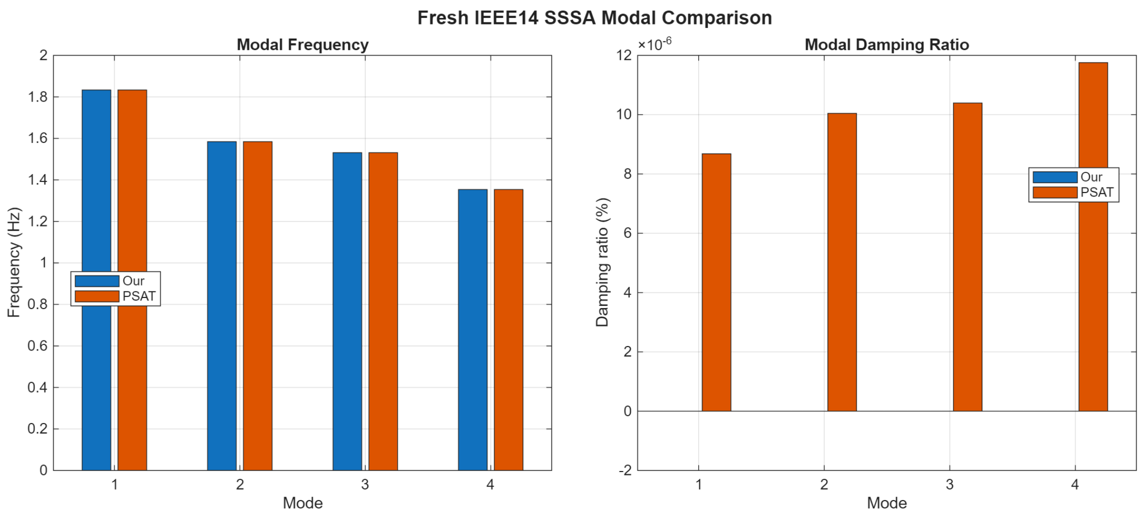

Figure 18: Fresh IEEE14 modal frequencies and damping ratios corresponding directly to Table 22.

8.5 Time-domain-simulation results

Generator columns are mapped by bus ID and direct trajectory differences are evaluated on the in-house event grid. The in-house run follows the same `solve_case('analysis','ts')` execution path and MATPOWER14 defaults used by `run_ts.m`. PSAT's additional internal event samples are retained in its raw count; interpolation to the comparison grid refuses extrapolation. Table 23 reports each tool's response summary and the direct PSAT–Our differences. All four figures plot the stored solver outputs: $\delta_i(t)$, $\omega_i(t)$, $P_{e,i}(t)$ and bus-voltage magnitude. No COI, reference-machine, initial-angle or speed-deviation transform is applied. In every figure the upper panel overlays Our as continuous lines and PSAT as cross markers sampled from the same comparison grid, making coincident trajectories visible. The lower panel plots the direct PSAT–Our difference.

Table 23: Fresh IEEE14 TDS comparison for the bus-4, $Z_f = j10^{-4}$ pu fault.

Metric	Unit	Our	PSAT
Completed to 15 s	–	Yes	Yes
Raw time points	points	1501	1509
Non-converged steps	steps	0	Not exposed
Minimum fault-bus voltage	pu	0.000973	0.000973
Maximum stored rotor-angle magnitude	deg	4964.684009	4964.511399
Maximum stored rotor-speed magnitude	pu	1.040163906	1.040155053
Maximum absolute electrical power	MW	300.383981	300.376323
Maximum PSAT–Our rotor-angle difference	deg	Reference	1.863392e-01
Maximum PSAT–Our rotor-speed difference	pu	Reference	9.867292e-06
Maximum PSAT–Our electrical-power difference	MW	Reference	1.225034e-01
Maximum PSAT–Our bus-4 voltage difference	pu	Reference	7.810489e-05

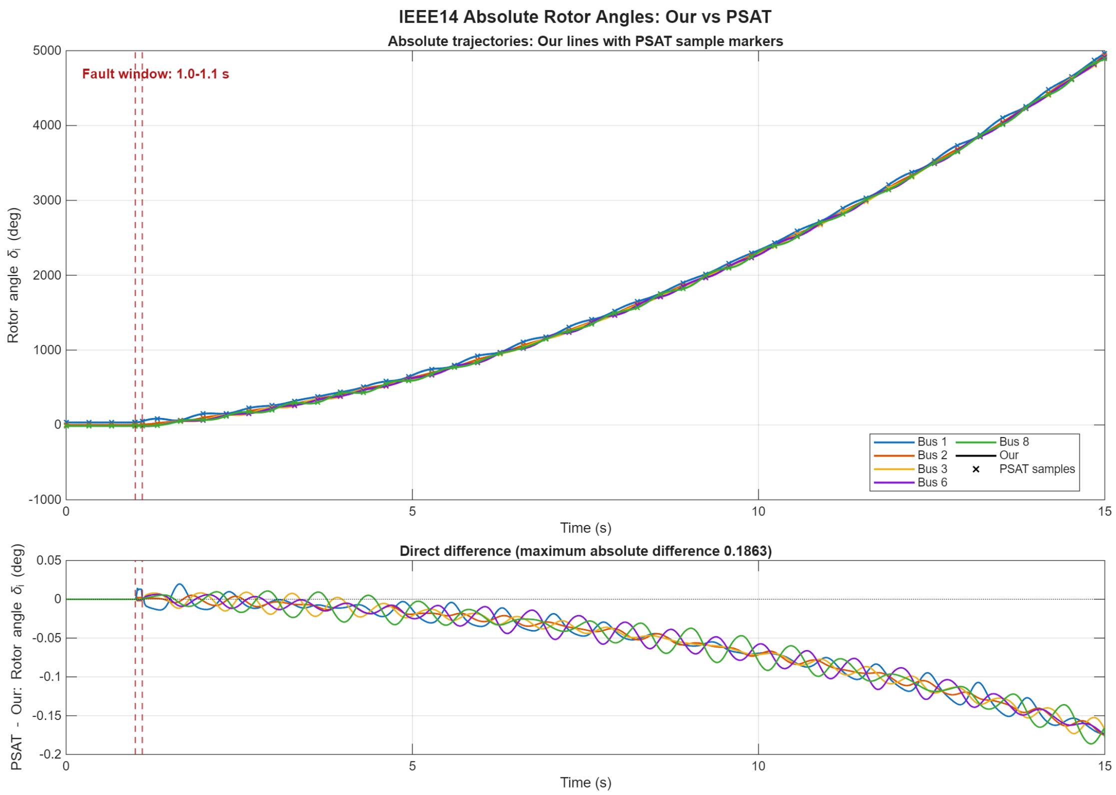

Figure 19: Fresh IEEE14 stored absolute rotor angles $\delta_i(t)$ for the five generator buses, using the `run_ts.m` plotting convention. No COI, reference-machine or initial-angle subtraction is applied. Colours identify buses. Upper panel: Our lines and PSAT cross markers. Lower panel: direct PSAT–Our angle differences.

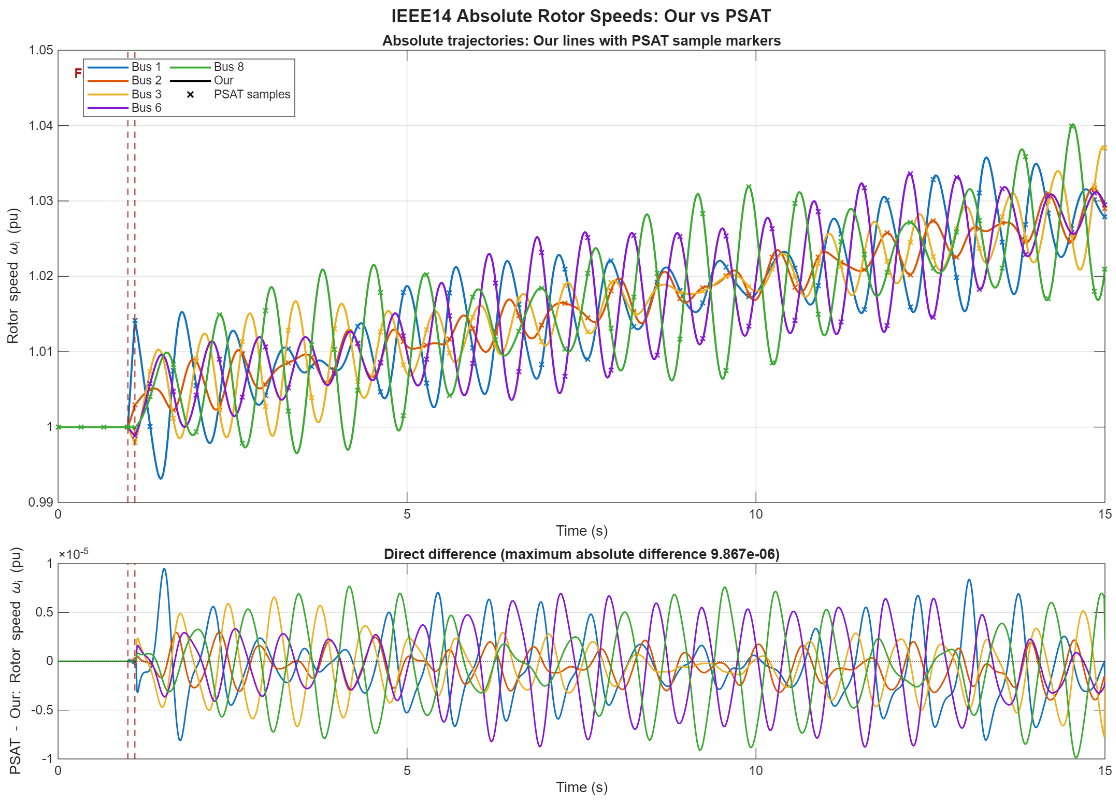

Figure 20: Fresh IEEE14 stored absolute rotor speeds $\omega_i(t)$ in per unit, using the `run_ts.m` plotting convention. No COI subtraction and no $\omega_i - 1$ transform is applied. Upper panel: Our lines and PSAT cross markers. Lower panel: direct PSAT–Our speed differences.

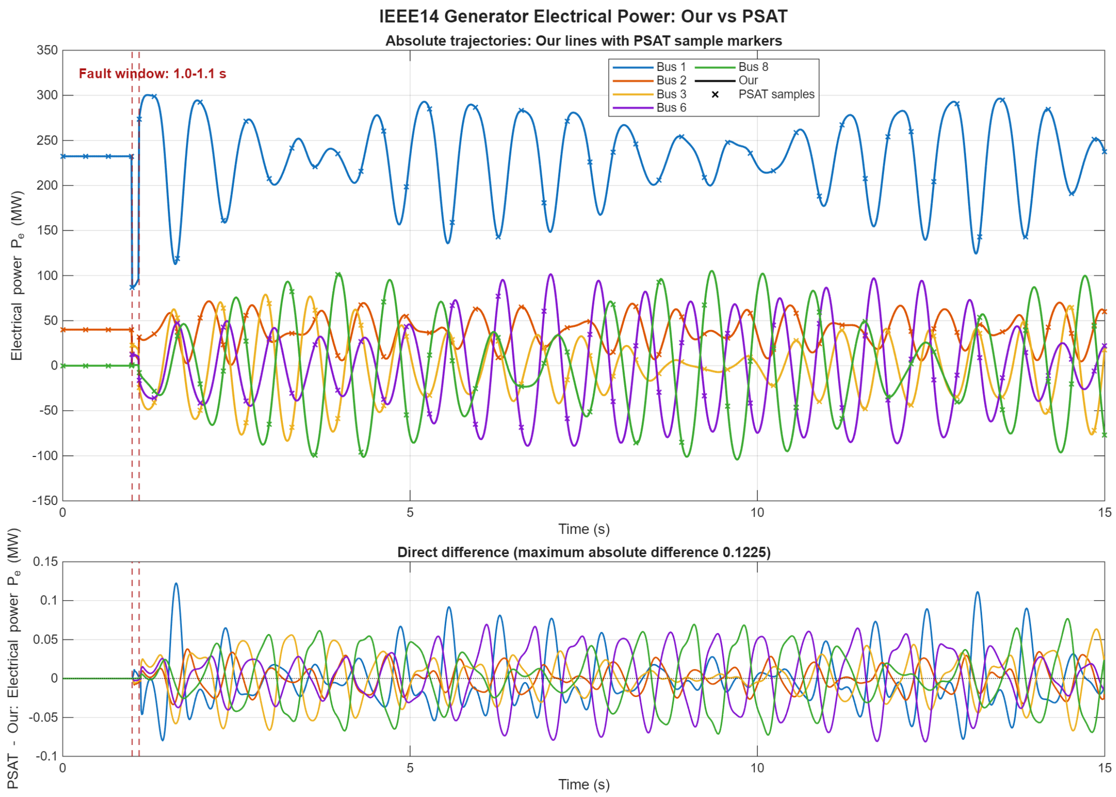

Figure 21: Fresh IEEE14 stored generator electrical powers $P_{e,i}(t)$, using the `run_ts.m` plotting convention. Upper panel: Our lines and PSAT cross markers. Lower panel: direct PSAT–Our electrical-power differences.

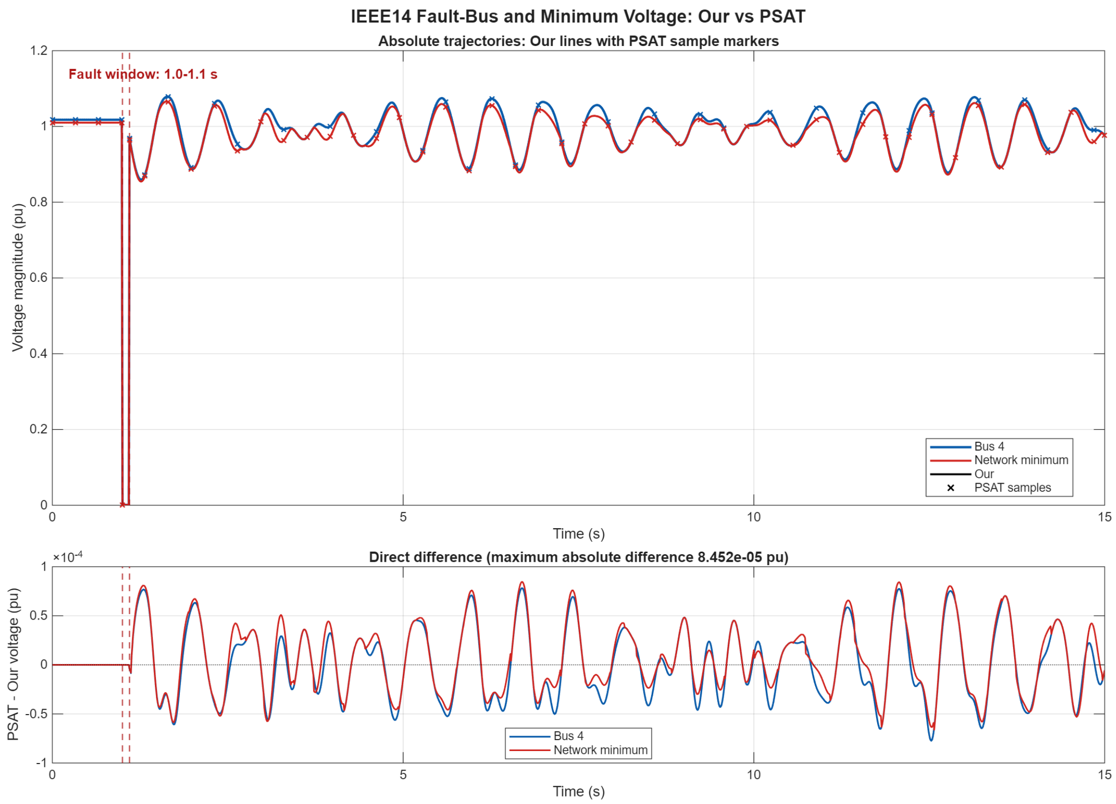

Figure 22: Fresh IEEE14 bus-4 and minimum-network voltage magnitudes for the declared fault, using the `run_ts.m` plotting convention. Upper panel: Our lines and PSAT cross markers. Lower panel: direct PSAT–Our voltage differences.

9 Conclusions

The Padiyar two-area case provides a traceable replacement reference for the project. It reproduces Table 9.2 with the in-house power flow, implements the source's actual five-state AVR model, exposes a manual-excitation comparison, and runs SSSA and residual-checked TS from one DAE. No subtransient parameter or PSS setting is fabricated. The next extension may add a documented PSS design and then use this frozen SG baseline for SG-to-VSG replacement studies.